

Estimation and Hypothesis Testing

Chapter 4 introduced likelihood and explored associated concepts such as likelihood ratio statistics and maximum likelihood estimators, which were then extensively used for inference in Chapters 5 and 6. In this chapter we turn aside from the central theme of the book and discuss some more theoretical topics. Estimation is a fundamental statistical activity, and in Section 7.1 we consider what properties a good estimator should have, including a brief discussion of nonparametric density estimators and the mathematically appealing topic of minimum variance unbiased estimation. One of the most important approaches to constructing estimators is as solutions to systems of estimating equations. In Section 7.2 we discuss the implications of this, showing how it complements minimum variance unbiased estimation, and seeing its implications for robust estimation and for stochastic processes. We then give an account of some of the main ideas underlying another major statistical activity, the testing of hypotheses, discussing the construction of tests with good properties, and making the connection to estimation.

7.1 Estimation

7.1.1 Mean squared error

Suppose that we wish to estimate some aspect of a probability model $f(y)$. In principle we might try and estimate almost any feature of f , but we largely confine ourselves to estimation of the unknown parameter θ or a function of it $\psi(\theta)$ in a parametric model $f(y; \theta)$. Suppose that our data Y comprise a random sample $Y_1, \dots, Y_n$ from f , and let the statistic $T = t(Y)$ be an estimator of $\psi(\theta)$. We say that T is *unbiased* for $\psi(\theta)$ if $E(T) = \psi(\theta)$ for all θ , and define the *bias* of T to be $E(T) - \psi(\theta)$. Large bias means that the long-run average value of T lies far from $\psi(\theta)$, and this is undesirable. The *mean squared error* of T is the expected squared distance between T and its estimand, which turns out to equal the sum of the variance and squared bias:

$$\begin{aligned} E\{[T - \psi(\theta)]^2\} &= E\{[T - E(T) + E(T) - \psi(\theta)]^2\} \\ &= \text{var}(T) + \{E(T) - \psi(\theta)\}^2, \end{aligned} \quad (7.1)$$

because $E[\{T - E(T)\}\{E(T) - \psi(\theta)\}] = 0$. Mean squared error is a common measure of how well T estimates $\psi(\theta)$. The decomposition (7.1) is useful because in practice it helps to know if a large mean squared error is due to a large bias or large variance or both.

Example 7.1 (Normal variance) Let $Y_1, \dots, Y_n$ be a random sample from the normal distribution with mean μ and variance σ^2 . Then σ^2 has maximum likelihood estimator $\hat{\sigma}^2 = n^{-1} \sum_j (Y_j - \bar{Y})^2 \stackrel{D}{=} n^{-1} \sigma^2 V$, where V has a χ_{n-1}^2 distribution, by (3.15). As $E(V) = (n-1)$ and $\text{var}(V) = 2(n-1)$,

$$E(\hat{\sigma}^2) = n^{-1}(n-1)\sigma^2, \quad \text{var}(\hat{\sigma}^2) = n^{-2}2(n-1)\sigma^4.$$

Hence $\hat{\sigma}^2$ has a negative bias of

$$E(\hat{\sigma}^2) - \sigma^2 = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}$$

and mean squared error

$$2\frac{n-1}{n^2}\sigma^4 + \left(-\frac{\sigma^2}{n}\right)^2 = \frac{2n-1}{n^2}\sigma^4.$$

The usual estimator is $S^2 = (n-1)^{-1} \sum_j (Y_j - \bar{Y})^2 \stackrel{D}{=} (n-1)^{-1} \sigma^2 V$, so

$$E(S^2) = (n-1)^{-1}\sigma^2(n-1) = \sigma^2, \quad \text{var}(S^2) = 2(n-1)\sigma^4(n-1)^{-2} = \frac{2}{n-1}\sigma^4.$$

Hence S^2 is unbiased, but its mean squared error is greater than that of $\hat{\sigma}^2$ because $(2n-1)/n^2 < 2/(n-1)$ for all $n > 1$. ■

Both bias and variance are $O(n^{-1})$ in this example, so at least for large n the contribution that bias squared makes to mean squared error is negligible compared to that of variance. This often occurs with parametric estimators, as the following argument suggests. Suppose that the data are a random sample from a distribution with mean μ , variance σ^2 , third cumulant κ_3 , and well-behaved higher cumulants. Then (2.32) implies that their average $\bar{Y}$ has mean μ , variance σ^2/n , and third cumulant $\kappa_3/n^{3/2}$. Suppose also that the estimator of a scalar parameter $\psi = t(\mu)$ may be written as $\hat{\psi} = t(\bar{Y})$, as is true for natural exponential family models, for example; see Section 5.2. Then under mild conditions, Taylor series expansion of $t(\bar{Y})$ about μ gives

$$\hat{\psi} = t(\mu) + (\bar{Y} - \mu)t'(\mu) + \frac{1}{2}(\bar{Y} - \mu)^2 t''(\mu) + \frac{1}{6}(\bar{Y} - \mu)^3 t'''(\mu) + \dots,$$

so

$$E(\hat{\psi}) = t(\mu) + \frac{1}{2}n^{-1}\sigma^2 t''(\mu) + \frac{1}{6}n^{-3/2}\kappa_3 t'''(\mu) + \dots,$$

showing that the bias of $\hat{\psi}$ is of order n^{-1} . Thus for asymptotic purposes we can typically compare two parametric estimators in terms of their variances, as squared bias is of smaller asymptotic order. In finite samples the issue is less clear, because it

is not useful to be told that $E(T) - \psi \doteq a/n$ and $\text{var}(T) \doteq b/n$ without some idea of a and b : if $a^2/n^2 \gg b/n$ for all values of n likely to be met in a particular application, then the bias term in (7.1) predominates and comparison purely in terms of variances is unhelpful. Nonetheless it is natural to ask if there is a lower bound to the variance of estimators of ψ .

Cramér–Rao lower bound

Suppose that the density $f(y; \theta)$ is regular for maximum likelihood estimation of the scalar parameter θ . If T is an unbiased estimator of a scalar $\psi = \psi(\theta)$, then under mild conditions and for all θ ,

$$\text{var}(T) \geq \frac{(d\psi/d\theta)^2}{I(\theta)}, \quad (7.2)$$

where $I(\theta)$ is the expected information in the sample. The right-hand side of (7.2) is the *Cramér–Rao lower bound*. It follows that if we can find an unbiased estimator that attains equality or near-equality in (7.2), we need search no further: no unbiased estimator could do better. We might also hope that when T has a small bias and its variance is close to the lower bound, it will be difficult to ameliorate.

To establish (7.2), note that as T is unbiased and $f(y; \theta)$ is a density,

$$E(T) = \int t(y) f(y; \theta) dy = \psi(\theta), \quad \int f(y; \theta) dy = 1,$$

for all θ . If the order of integration and differentiation can be interchanged, differentiation of these equations with respect to θ gives

$$\frac{d\psi}{d\theta} = \int t(y) \frac{df(y; \theta)}{d\theta} dy = \int t(y) \frac{d \log f(y; \theta)}{d\theta} f(y; \theta) dy = E(TU),$$

where $U = d \log f(Y; \theta)/d\theta$ is the score statistic, and

$$0 = \int \frac{df(y; \theta)}{d\theta} dy = \int \frac{d \log f(y; \theta)}{d\theta} f(y; \theta) dy = E(U).$$

Hence $\text{cov}(T, U) = E(TU) - E(T)E(U) = d\psi/d\theta$. Moreover

$$\text{var}(U) = E(U^2) = E \left\{ -\frac{d^2 \log f(Y; \theta)}{d\theta^2} \right\} = I(\theta)$$

by (4.33). But the Cauchy–Schwarz inequality (Exercise 2.2.3) gives

$$\text{cov}(T, U)^2 \leq \text{var}(T)\text{var}(U),$$

or $(d\psi/d\theta)^2 \leq \text{var}(T)I(\theta)$, which entails (7.2). This will apply if maximum likelihood estimation is regular, for example. Equality in (7.2) only occurs when there is linear dependence between T and U , so that $T = b_1(\theta) + b_2(\theta)U$ for all θ and some constants $b_1(\theta)$ and $b_2(\theta) \neq 0$.

The inverse Fisher information for ψ is $I(\psi)^{-1} = (d\psi/d\theta)^{-2}I(\theta)^{-1}$, which equals the right-hand side of (7.2) (Problem 4.2). Hence the Cramér–Rao lower bound for estimation of ψ is $I(\psi)^{-1}$, for any sample size. However (4.26) implies that in regular cases, the large-sample distribution of the maximum likelihood estimator, $\hat{\psi}$, is normal

Harald Cramér (1893–1985) worked all his life at the University of Stockholm, researching in biochemistry and number theory before turning to probability and statistics. He was the first Swedish professor of mathematical statistics and actuarial science, and made fundamental contributions to both subjects. His masterpiece, *Mathematical Methods of Statistics*, was written during World War II, when he was largely scientifically isolated. Calyampudi Radhakrishnan Rao (1920–) works at Pennsylvania State University. See DeGroot (1987b) for an account of his life and work.

Table 7.1 Cramér–Rao lower bound for estimation of mean of a log-normal sample of size n , when $\mu = 0$ and $\sigma = 1.5$, with properties of estimators $T = \bar{Y}$ and $\hat{\psi} = \exp(\hat{\sigma}^2/2)$.

n	5	10	20	40	80	160	320
Cramér–Rao lower bound	4.80	2.40	1.20	0.60	0.30	0.15	0.08
$\text{var}(T)$	16.11	8.05	4.03	2.01	1.01	0.50	0.25
$\text{var}(\hat{\psi})$	296.36	7.08	1.91	0.75	0.33	0.16	0.08
Bias of $\hat{\psi}$ ($\times 10^{-2}$)	137.73	49.66	21.83	10.30	5.01	2.47	1.23
Mean squared error of $\hat{\psi}$	298.26	7.32	1.96	0.76	0.34	0.16	0.08

with mean ψ and variance $I(\psi)^{-1}$. Thus $\hat{\psi}$ is asymptotically unbiased and attains the Cramér–Rao lower bound as $n \rightarrow \infty$: it has asymptotically the smallest possible variance among unbiased estimators. Such an estimator is said to be *efficient*. This suggests that the *asymptotic relative efficiency* of T be defined as

$$\frac{\text{var}(\hat{\psi})}{\text{var}(T)} = \frac{(d\psi/d\theta)^2}{I(\theta)\text{var}(T)},$$

generalizing (4.19).

In regular cases $\text{var}(\hat{\psi}) = O(n^{-1})$, where n is sample size, and it is impossible to find an unbiased estimator of ψ with variance of smaller order than this. If in addition the observations are identically distributed, the Cramér–Rao lower bound for θ is $1/\{ni(\psi)\}$, where $i(\psi)$ is the information in a single observation.

Example 7.2 (Log-normal mean) A log-normal random variable Y may be expressed as $\exp(\mu + \sigma Z)$, where Z is a standard normal variable; thus $X = \log Y$ has the $N(\mu, \sigma^2)$ distribution. The mean and variance of Y are (Problem 3.5)

$$\psi = e^{\mu + \sigma^2/2}, \quad e^{2\mu + \sigma^2}(e^{\sigma^2} - 1).$$

If it is known that $\mu = 0$, then the maximum likelihood estimator of σ^2 based on a random sample $Y_1, \dots, Y_n$ is $\hat{\sigma}_0^2 = n^{-1} \sum X_j^2$, where $X_j = \log Y_j$, and two possible estimators for ψ are $T = \bar{Y}$ and $\hat{\psi} = \exp(\hat{\sigma}_0^2/2)$. Now $\hat{\sigma}_0^2 \stackrel{D}{=} \sigma^2 V/n$, where V has a chi-squared distribution on n degrees of freedom, so

$$E(\hat{\psi}^r) = E\{\exp(r\sigma^2 V/2n)\} = (1 - r\sigma^2/n)^{-n/2}, \quad n = 1, 2, \dots,$$

because V has moment-generating function $(1 - 2t)^{-n/2}$. This enables exact calculation of the bias, variance and mean squared error of $\hat{\psi}$.

Evidently T is an unbiased estimator of ψ with variance and mean squared error both equal to $n^{-1} \exp(\sigma^2)\{\exp(\sigma^2) - 1\}$. As $I(\sigma^2) = \sigma^4/(2n)$ and $d\psi/d\sigma^2 = \frac{1}{2}\psi$, the Cramér–Rao lower bound for ψ is $\sigma^4 \exp(\sigma^2)/(2n)$.

The first three lines of Table 7.1 give values of the lower bound, $\text{var}(T)$ and $\text{var}(\hat{\psi})$ for various n when $\sigma = 1.5$, so $\psi = \exp(\sigma^2/2) = 9.49$. For $n \geq 80$ the bound is effectively attained by $\hat{\psi}$, whose bias is always small compared to its variance. For $n < 80$ the bias and variance of $\hat{\psi}$ decrease rather faster than their asymptotic rate n^{-1} . Even when $n = 5$ the contribution to mean squared error from bias is very small. The unbiased estimator T is much more efficient when $n = 5$, but otherwise is beaten

by $\hat{\psi}$; its asymptotic relative efficiency is

$$\frac{\text{var}(\hat{\psi})}{\text{var}(T)} = \frac{\sigma^4 e^{\sigma^2}}{2e^{\sigma^2}(e^{\sigma^2} - 1)} = \frac{\sigma^4}{2(e^{\sigma^2} - 1)} \doteq 0.3$$

when $\sigma = 1.5$, close to the variance ratio in the last columns. ■

Example 7.3 (Uniform distribution) Let $Y_1, \dots, Y_n$ be a random sample from the uniform distribution on $(0, \theta)$. The likelihood for θ is

$$L(\theta) = \begin{cases} \theta^{-n}, & 0 \leq Y_1, \dots, Y_n \leq \theta, \\ 0, & \text{otherwise,} \end{cases}$$

or equivalently and in terms of the largest order statistic $Y_{(n)}$,

$$L(\theta) = \begin{cases} \theta^{-n}, & 0 \leq Y_{(n)} \leq \theta, \\ 0, & \text{otherwise,} \end{cases}$$

a sketch of which shows that $\hat{\theta} = Y_{(n)}$. Hence $\hat{\theta}$ has distribution

$$\Pr(\hat{\theta} \leq u) = \begin{cases} 0, & u < 0, \\ (u/\theta)^n, & 0 \leq u < \theta, \\ 1, & u \geq \theta, \end{cases}$$

and it is straightforward to see that

$$E(\hat{\theta}) = \frac{n\theta}{n+1}, \quad \text{var}(\hat{\theta}) = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

This variance is $O(n^{-2})$, suggesting a potential problem with the Cramér–Rao lower bound. In fact in this case

$$\frac{d}{d\theta} \int t(y)f(y; \theta) dy = \frac{d}{d\theta} \int_0^\theta \frac{t(y)}{\theta} dy = \frac{t(\theta)}{\theta} - \int_0^\theta \frac{t(y)}{\theta^2} dy \neq \int t(y) \frac{df(y; \theta)}{d\theta} dy,$$

because the limit of the integral depends on θ . Here the model is non-regular and the lower bound does not apply. ■

The lower bound extends to the case where θ has dimension p and $\psi = \psi(\theta)$ has dimension $q \leq p$. Suppose that the $q \times 1$ statistic T is an unbiased estimator of ψ , with $q \times q$ covariance matrix $\text{var}(T)$. Then $E(T) = \psi(\theta)$ for all θ , so $\partial\psi/\partial\theta^T$ is a $q \times p$ matrix. An argument analogous to that on page 302 shows that the $q \times q$ matrix

$$\text{var}(T) - \frac{\partial\psi}{\partial\theta^T} I(\theta)^{-1} \frac{\partial\psi}{\partial\theta}$$

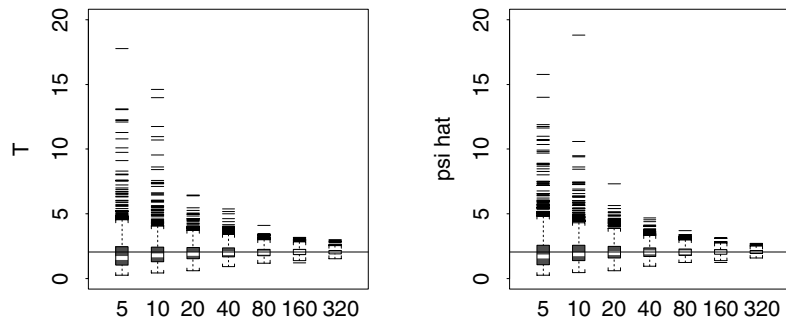
is positive semi-definite. If ψ is scalar, it follows that

$$\text{var}(T) \geq \frac{\partial\psi}{\partial\theta^T} I(\theta)^{-1} \frac{\partial\psi}{\partial\theta}, \quad (7.3)$$

which extends (7.2).

Example 7.4 (Log-normal mean) Expression (4.18) implies that the inverse Fisher information for the parameters of a normal distribution with parameters $\theta = (\mu, \sigma^2)^T$

Figure 7.1 Unbiased (left) and maximum likelihood estimates (right) of log-normal mean based on samples of size n when $\mu = 0$, $\sigma = 1.5$. The horizontal lines show the target parameter ψ . The white band in the centre of the boxplots indicates the median of the simulated values.



based on a random sample $X_1, \dots, X_n$ is

$$I(\mu, \sigma^2)^{-1} = \begin{pmatrix} \sigma^2/n & 0 \\ 0 & 2\sigma^4/n \end{pmatrix}.$$

The maximum likelihood estimators are $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_j - \bar{X})^2$. Let $\psi = \exp(\mu + \sigma^2/2)$ denote the mean of the log-normal variable $Y = \exp(X)$. Then

$$\frac{\partial \psi}{\partial \theta} = \begin{pmatrix} e^{\mu + \sigma^2/2} \\ \frac{1}{2} e^{\mu + \sigma^2/2} \end{pmatrix}$$

and the Cramér–Rao lower bound (7.3) for ψ is $n^{-1}(\sigma^2 + \sigma^4/2)e^{2\mu + \sigma^2}$; this is also the asymptotic variance of the maximum likelihood estimator $\hat{\psi} = \exp(\hat{\mu} + \hat{\sigma}^2/2)$. As in Example 7.2, the exact bias and variance of $\hat{\psi}$ may be obtained explicitly (Problem 7.1).

Figure 7.1 shows 1000 simulated values of T and $\hat{\psi}$ for various n . Their appreciable skewness when n is small shows that their distributions are then far from normal and calls into question our use of bias and variance to compare them. As $\psi > 0$, a measure of relative error such as $E\{(T - \psi)^2\}/\psi^2$ might be preferable. ■

Mean squared error is a useful measure with which to compare estimators, but it has the disadvantage of being tied to a particular scale. Thus even if T is preferable to T' as an estimator of ψ , $g(T)$ need not be better than $g(T')$ when estimating $g(\psi)$. In particular, if T is unbiased, then $g(T)$ is unbiased only when g is linear. This is not critical when interest focuses on a quantity for which transformations are irrelevant, but is awkward otherwise.

7.1.2 Kernel density estimation

The examples above suggest that estimators in parametric problems can often be compared in terms of their variances, bias being relatively unimportant. Bias and variance play more balanced roles in other contexts, as we now illustrate with a discussion of nonparametric density estimation. We also take the opportunity of introducing cross-validation, which plays a role later.

The elementary estimator of an unknown density $f(y)$, the histogram, has a number of drawbacks. In addition to often looking rather rough due to the use of bins with

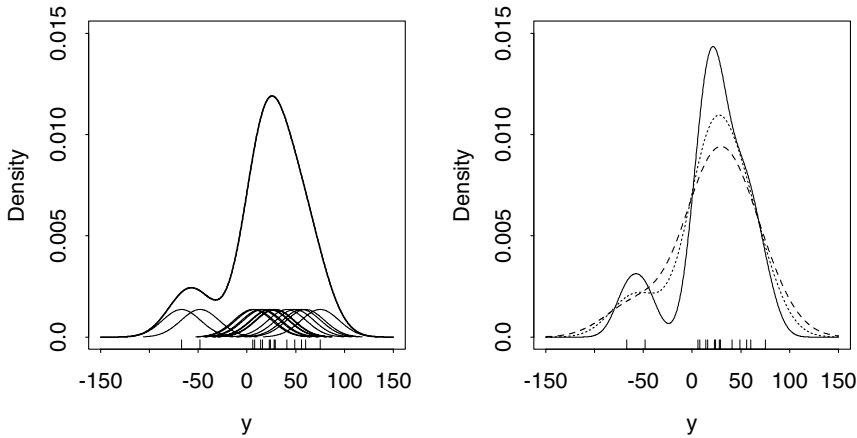


Figure 7.2 Kernel density estimates for maize data. Left: construction of kernel estimate (heavy) as sum of 15 scaled normal densities centred at the y_j , with $h = 19.5$. Right: density estimates with $h = 13.3$ (solid), $h = 23.2$ (dots) and $h = 30$ (dashes).

sharp edges, its appearance depends heavily on the placing of the bin boundaries and on bin width. Slight changes to the boundaries can give strikingly different histograms for the same data, particularly with small samples, and this is clearly undesirable.

An alternative approach is based on a kernel function $w(y)$, which is a symmetric probability density with mean zero and unit variance, descending smoothly to zero as $|y| \rightarrow \infty$. An example is the standard normal density, but there are many other possibilities. The *kernel density estimator* based on a random sample $Y_1, \dots, Y_n$ from f is

$$\hat{f}_h(y) = \frac{1}{nh} \sum_{j=1}^n w\left(\frac{y - Y_j}{h}\right), \quad (7.4)$$

where $h > 0$ is a bandwidth. Figure 7.2 shows how this is constructed for the data in the rightmost column of Table 1.1: the value of $\hat{f}_h$ at y is obtained by summing contributions from densities with standard deviations h and centred at each of the Y_j , with those closest to y contributing most. The estimate depends on both bandwidth and kernel, but the choice of h is much the more important. When n is large, a smaller bandwidth can be chosen, so that the contributions are more localized. Such estimators are commonly available in statistical packages and are widely used in applications.

To find the mean and variance of $\hat{f}_h(y)$, note that

$$\mathbb{E} \left\{ \frac{1}{nh} \sum_{j=1}^n w\left(\frac{y - Y_j}{h}\right) \right\} = \mathbb{E} \left\{ h^{-1} w\left(\frac{y - Y}{h}\right) \right\}.$$

Symmetry of the kernel gives

$$h^{-1} \int w\left(\frac{y-x}{h}\right) f(x) dx = h^{-1} \int w\left(\frac{x-y}{h}\right) f(x) dx = \int w(u) f(y+hu) du,$$

and Taylor series expansion for small h gives

$$\int w(u) \left\{ f(y) + hu f'(y) + \frac{1}{2} h^2 u^2 f''(y) + \dots \right\} du = f(y) + \frac{1}{2} h^2 f''(y) + O(h^4), \quad (7.5)$$

Here $f'(y) = df(y)/dy$, and so forth.

because of the symmetry and moment properties of w . Thus $\hat{f}_h(y)$ has bias $\frac{1}{2}h^2 f''(y) + O(h^4)$. The variance of $\hat{f}_h(y)$ is

$$\begin{aligned}\text{var}\{\hat{f}_h(y)\} &= \frac{1}{nh^2} \text{var} \left\{ w \left(\frac{Y-y}{h} \right) \right\} \\ &= \frac{1}{nh^2} \left[E \left\{ w \left(\frac{Y-y}{h} \right)^2 \right\} - E \left\{ w \left(\frac{Y-y}{h} \right) \right\}^2 \right],\end{aligned}$$

which depends on

$$E \left\{ h^{-2} w \left(\frac{Y-y}{h} \right)^2 \right\}, \quad E \left\{ h^{-1} w \left(\frac{Y-y}{h} \right) \right\}.$$

An argument similar to that above shows that the first of these is

$$h^{-1} \int w(u)^2 \{f(y) + huf'(y) + \dots\} du = h^{-1} f(y) \int w(u)^2 du + O(h), \quad (7.6)$$

while we have already seen that the second is $O(1)$. Thus as $h \rightarrow 0$, the variance of $\hat{f}_h(y)$ is of order $(nh)^{-1}$.

If both the bias obtained from (7.5) and the variance obtained from (7.6) are to vanish as $n \rightarrow \infty$ we must choose h so that $h \rightarrow 0$ and $nh \rightarrow \infty$, for example taking $h \propto n^{-1/2}$. But does this give the best bias-variance tradeoff? The leading terms of the asymptotic mean squared error of $\hat{f}_h(y)$ are

$$\frac{h^4}{4} f''(y)^2 + \frac{1}{nh} f(y) \int w(u)^2 du. \quad (7.7)$$

Differentiation shows that this is minimized as a function of h at

$$h_{opt}(y, f) = n^{-1/5} \left\{ \int w(u)^2 du f(y)/f''(y)^2 \right\}^{1/5},$$

giving an optimal bandwidth that varies with y . Choosing $h \propto n^{-1/5}$ gives bias of order $n^{-2/5}$ and variance of order $n^{-4/5}$, and then bias squared and variance contribute terms of equal order to (7.7). As far as variance is concerned, the effective local sample size is $n^{-4/5}$, so if $n = 1000$, say, the number of observations contributing to a local estimator will be roughly 250, considerably fewer than the sample size applicable for parametric estimation; this is typical of local problems, for which larger samples are needed.

The bandwidth plays a key role in kernel density estimation. In exploratory work a good strategy is to try several values, in the hope that different amounts of smoothing will reveal different interesting features of the data. However automatic rules for selection of h can be valuable, both to suggest starting-points for exploration and for use when density estimation plays a minor role in a larger analysis. The key difficulty is that as we saw above, an optimal choice such as $h_{opt}(y, f)$ depends on the unknown f as well as on y . Dependence on y can be removed by minimizing an overall measure

of the distance between $\hat{f}_h$ and f , such as the integral of (7.7) over y ,

$$\frac{h^4}{4} \int f''(y)^2 dy + \frac{1}{nh} \int w(u)^2 du, \quad (7.8)$$

but the dependence on f remains. Many proposals have been made to sidestep it, of which we outline only two.

One simple approach is to find the bandwidth that would be optimal if f were known. If it is normal with variance σ^2 , for example, and $w(u) = \phi(u)$, then the choice minimizing (7.8) is $h = 1.06\sigma n^{-1/5}$. In practice σ is replaced by its sample counterpart s , but as this is sensitive to outliers it may be better to use a rule of thumb such as

$$h'_{opt} = 0.9 \min \{s, \text{IQR}/1.35\} n^{-1/5}.$$

IQR is the sample interquartile range.

This choice is very simple but tends to oversmooth for non-normal data, thereby obscuring multimodality and other features of potential interest.

A better approach is to minimize an estimate of the exact mean integrated squared error of $\hat{f}_h(y)$,

$$E \int \{\hat{f}_h(y) - f(y)\}^2 dy = E \int \hat{f}_h(y)^2 dy - 2E \int \hat{f}_h(y)f(y) dy + \int f(y)^2 dy, \quad (7.9)$$

whose true value involves expectation with respect to f and so is unknown. The third term of this does not depend on h , so it suffices to estimate the first two terms. The awkward term is the second, an obvious estimator of which may be expressed in terms of the empirical distribution function $\hat{f}$ as

$$\int \hat{f}_h(y) d\hat{f}(y) = \frac{1}{n} \sum_{j=1}^n \hat{f}_h(y_j),$$

but this is biased because y_j appears twice in $\hat{f}_h(y_j)$, once implicitly. The bias can be removed by using the *cross-validation estimator* given by

$$\frac{1}{n} \sum_{j=1}^n \hat{f}_{h,-j}(y_j), \quad \text{where} \quad \hat{f}_{h,-j}(y) = \frac{1}{(n-1)h} \sum_{i \neq j} w\left(\frac{y - y_i}{h}\right)$$

is based on all observations except y_j and so is called a *leave-one-out* estimator. The term cross-validation means that each datum is compared with the rest. Cross-validation is widely used and has many variants. It is related to AIC; in both cases the complexity of a set of models would ideally be assessed using the accuracy of their predictions for data sets like the original. As a new dataset is unavailable, we manufacture one with just one datum, assess how well that datum can be predicted using the remainder, and average all n possible such comparisons.

It is easily shown that apart from a constant factor the first two terms of (7.9) are estimated unbiasedly by

$$CV(h) = \frac{1}{n} \sum_{j=1}^n \int \hat{f}_{h,-j}(y)^2 dy - \frac{2}{n} \sum_{j=1}^n \hat{f}_{h,-j}(y_j),$$

so it seems reasonable to hope that the value $\hat{h}_{opt}^{CV}$ that minimizes this will be close to the value $h_{opt}(f)$ which minimizes (7.9). If the kernel is normal, then $CV(h)$ is readily computed using results on convolutions of normal densities. Care is needed when minimizing $CV(h)$, as it may have several local minima.

Although close to unbiased, $\hat{h}_{opt}^{CV}$ is rather variable. Other formulations of cross-validation can be used to derive more stable bandwidth estimators, but typically at the expense of increased bias. Section 7.4 gives further reading.

Example 7.5 (Maize data) The maize data have $IQR = 34$ and $s = 37.74$, so $1.06sn^{-1/5} = 23.2$ and $h'_{opt} = 13.3$. The effect of these choices may be seen in Figure 7.2, where using h'_{opt} makes the two negative observations more prominent, perhaps too much so, and $h = 30$ seems rather large. Cross-validation gives $\hat{h}_{opt}^{CV} = 19.5$, again over-emphasizing the two negative y_j .

Here $n = 15$ and it is unreasonable to hope for a useful nonparametric density estimate. ■

7.1.3 Minimum variance unbiased estimation

Other things being equal, bias is undesirable. A badly biased estimator has expected value far from the parameter it is supposed to estimate, so it seems a good idea to minimize bias so far as possible. This has motivated a careful study of unbiased estimators, for which there is a rather complete theory in a limited class of models.

Rao–Blackwell theorem

Suppose that data Y arise from a statistical model $f(y; \theta)$ and that we want to estimate a scalar function $\psi = \psi(\theta)$ of θ . Suppose also that the statistic $S = s(Y)$ is sufficient for θ , and that $T = t(Y)$ is an unbiased estimator of ψ . Then subject to suitable regularity conditions, $W = w(S) = E(T | S)$ is an unbiased estimator of ψ with variance no larger than that of T :

$$\text{var}(T) \geq \text{var}(W) \quad \text{for all } \theta. \quad (7.10)$$

This is the *Rao–Blackwell theorem*. It is a non-asymptotic result, applicable to samples of any size.

To establish (7.10), first note that W is indeed a statistic:

$$W = w(S) = E(T | S) = \int t(y)f(y | s)dy$$

does not depend on θ because of the sufficiency of S . Therefore W can be calculated from Y alone. Secondly,

$$E(W) = E_S\{E(T | S)\} = E(T) = \psi,$$

so W is unbiased for ψ . Thirdly,

$$\begin{aligned} \text{var}(T) &= E\{(T - \psi)^2\} \\ &= E[\{T - E(T | S) + E(T | S) - \psi\}^2] \end{aligned}$$

David Harold Blackwell (1919–) works at the University of California, Berkeley. See DeGroot (1986a) for an account of his life and work.

$$\begin{aligned}
&= E[\{T - E(T | S)\}^2] + 2E\{(T - W)(W - \psi)\} + E\{(W - \psi)^2\} \\
&= E\{(T - W)^2\} + \text{var}(W),
\end{aligned} \tag{7.11}$$

because the middle term is

$$\begin{aligned}
E_Y\{(T - W)(W - \psi)\} &= E_S E_{Y|S}\{(T - W)(W - \psi)\} \\
&= E_S\{(W - W)(W - \psi)\} \\
&= 0.
\end{aligned}$$

Evidently (7.11) implies (7.10), giving $E\{(T - W)^2\} = 0$, that is, $T = W$ with probability one, so T and W are effectively the same estimator. The process of replacing an unbiased estimator T with another $E(T | S)$ with smaller variance is called *Rao–Blackwellization*.

Example 7.6 (Poisson mean) Let $Y_1, \dots, Y_n$ be a Poisson random sample whose mean θ we intend to estimate. Now $E(Y_1) = \theta$, so Y_1 is an unbiased estimator of θ . As the Poisson density is an exponential family, $S = \sum Y_j$ is minimal sufficient for θ (Section 5.2.3). Therefore $W = E(Y_1 | S)$ is an unbiased estimator for θ with variance at most that of Y_1 .

To find W , we argue by symmetry. Evidently $E(Y_1 | S) = \dots = E(Y_n | S)$, because the Y_j were independent and identically distributed unconditionally and the conditioning statistic S is symmetric in the Y_j . Therefore

$$E(Y_1 | S) = n^{-1} \sum_{j=1}^n E(Y_j | S) = E\left(n^{-1} \sum_{j=1}^n Y_j \middle| S\right) = E(n^{-1} S | S) = S/n.$$

This estimator has variance θ/n , whereas Y_1 has variance θ . ■

Example 7.7 (Dirac comb) A random sample of pairs (X, Y) from a bivariate density $f(x, y)$ is available, and we wish to estimate some feature of the unknown marginal density $f(y)$ of Y . Suppose that the conditional density $f(y | x)$ is available. This may seem unrealistic, but in fact this situation often arises when using simulation to estimate a density; see Section 11.3.3.

The likelihood based on data $(x_1, y_1), \dots, (x_n, y_n)$ is

$$\prod_{j=1}^n f(y_j | x_j) f(x_j),$$

and as $f(y | x)$ is known, $X_1, \dots, X_n$ is sufficient for $f(x, y)$ and hence for $f(y)$.

Suppose that we wish to estimate $f(y)$ itself. We might use a kernel density estimator (7.4) with bandwidth $h > 0$, but this would be biased. We can remove the bias by letting $h \rightarrow 0$, thus obtaining the Dirac comb

$$\hat{f}(y) = \frac{1}{n} \sum_{j=1}^n \delta(y - Y_j).$$

The comb, which places an infinite spike at each Y_j and can be regarded as the derivative of the empirical distribution function (2.3), gives terrible estimates of $f(y)$,

Paul Adrien Maurice Dirac (1902–1984) was born in Bristol and studied there and at Cambridge, where he later held the professorship of mathematics once held by Newton. His unifying work on the basis of quantum mechanics and relativity led to his receiving the 1933 Nobel Prize in physics. An intensely private man, he had to be persuaded that the publicity would be greater if he refused it, as he originally intended.

$\delta(u)$ is the Dirac delta function.

but it is unbiased because

$$\int \delta(y - u)g(u) du = g(y)$$

for any function g . To apply the Rao–Blackwell theorem, note that

$$E(\hat{f}(y) \mid X_1, \dots, X_n) = \frac{1}{n} \sum_{j=1}^n \int \delta(y - y')f(y' \mid X_j) dy' = \frac{1}{n} \sum_{j=1}^n f(y \mid X_j). \quad (7.12)$$

This is a kernel estimator for which the kernel is the conditional density of Y given X ; it is much smoother than the comb. As the first two moments of $f(y \mid X_j)$ are

$$\begin{aligned} E_X \{f(y \mid X)\} &= \int f(y \mid x)f(x) dx = f(y), \\ E_X \{f(y \mid X)^2\} &= \int f(y \mid x)^2 f(x) dx = \int f(y \mid x)f(x, y) dx, \end{aligned} \quad (7.13)$$

we see that (7.12) is unbiased, with finite variance when (7.13) is finite, as will usually be the case.

A similar argument applies to the cumulative distribution function of Y and may be preferred by those wary of delta functions (Exercise 7.1.8). ■

Completeness

Given an unbiased estimator T of ψ and a sufficient statistic, S , the Rao–Blackwell theorem enables us to find an unbiased estimator with variance at most that of T . However there may be many unbiased estimators, each of which could be Rao–Blackwellized. Is there one with lowest variance?

To answer this question we need the notion of *completeness*. A statistic S is *complete* if for any function h ,

$$E\{h(S)\} = 0 \quad \text{for all } \theta \text{ implies that } h \equiv 0, \quad (7.14)$$

and is *boundedly complete* if (7.14) is true provided h is bounded. Evidently a complete statistic is also boundedly complete. If S is complete, we say that its density $f(s; \theta)$ is complete. As completeness must hold for all θ , it is a property of a family of densities rather than of a single density.

Completeness of minimal sufficient statistics is used to establish uniqueness of minimum variance unbiased estimators. Note the qualification here: sufficient statistics that are not minimal are not in general complete.

Example 7.8 (Poisson density) Suppose that Y is Poisson with mean $\theta > 0$, and that $h(Y)$ satisfies $E\{h(Y)\} = 0$ for every value of θ . Then its expectation is proportional to a power series which is identically zero on the positive half-line:

$$0 = E\{h(Y)\} = \sum_{y=0}^{\infty} h(y) \frac{\theta^y}{y!} e^{-\theta} \propto \sum_{y=0}^{\infty} \theta^y \frac{h(y)}{y!}, \quad \theta > 0.$$

Hence $h(0) = h(1) = \dots = 0$, and Y is complete.

Strictly speaking $h = 0$ almost everywhere with respect to the density of S .

Now consider a Poisson sample of size n . Then $S = (Y_1, \dots, Y_n)$ is sufficient for θ , and $h(S) = Y_1 - Y_2$ has expectation zero for all θ . This does not imply that $Y_1 = Y_2$, however, so S is not complete. The corresponding minimal sufficient statistic $\sum Y_j$ has a Poisson density, and is complete. ■

Example 7.9 (Uniform density) Suppose that Y is uniformly distributed on $(-\theta, \theta)$. Then $E(Y) = 0$ for every $\theta > 0$, but as $h(y) = y$ is not identically zero, Y is not complete. ■

Example 7.10 (Exponential family) Suppose that Y belongs to an exponential family of order p ,

$$f(y; \omega) = \exp\{s(y)^T \theta - \kappa(\theta)\} f_0(y), \quad y \in \mathcal{Y}, \theta \in \mathcal{N}.$$

If Y is continuous and $E\{h(Y)\} = 0$, then provided that $\mathcal{N}$ contains an open set around the origin,

$$E\{h(Y)\} = \int h(y) \exp\{s(y)^T \theta - \kappa(\theta)\} f_0(y) dy = 0$$

is proportional to the Laplace transform of $h(y) f_0(y)$. Then the uniqueness of Laplace transforms implies that $h(y) f_0(y) = 0$ except on sets of measure zero and thus $h(Y) \equiv 0$: Y is complete. When Y is discrete the corresponding argument involves series or polynomials, as in Example 7.8.

The same argument applies to any subfamily whose parameter space contains an open set around the origin, and in particular to all the standard exponential family models. ■

To see how completeness is used, suppose that we have a parametric model $f(y; \theta)$ with complete minimal sufficient statistic S , and two unbiased estimators of $\psi = \psi(\theta)$, namely $T = t(Y)$ and $T' = t'(Y)$. Let $W = E(T | S)$ and $W' = E(T' | S)$. Now $E(W - W') = 0$ for all θ , and both W and W' are functions of the data only through S . But S is complete, so $W = W'$ except on sets of measure zero, that is, W and W' are identical for all practical purposes. Thus Rao–Blackwellization of an unbiased estimator using a complete sufficient statistic always leads to W , and no unbiased estimator of ψ has smaller variance. For suppose T' is an unbiased estimator of ψ with smaller variance than W . Then by the Rao–Blackwell theorem, $W' = E(T' | S)$ satisfies

$$\text{var}(W') \leq \text{var}(T') < \text{var}(W),$$

which is impossible because $W' \equiv W$.

Example 7.11 (Normal density) Let $Y_1, \dots, Y_n$ be a $N(\mu, \sigma^2)$ random sample, where $n \geq 2$. We saw in Example 5.14 that $S = (\bar{Y}, \sum (Y_j - \bar{Y})^2)$ is minimal sufficient, and as its density is an exponential family of order 2 in which we can take $\Omega = (-\infty, \infty) \times (0, \infty)$, S is complete.

Now $\bar{Y}$ is an unbiased estimator of μ that is a function of S , and therefore it is the minimum variance unbiased estimator of μ . Likewise the minimum variance unbiased estimator of σ^2 is $(n-1)^{-1} \sum (Y_j - \bar{Y})^2$. ■

Although of theoretical interest, minimum variance unbiased estimators are not widely used in practice. One difficulty is that the restriction to exact unbiasedness can exclude every interesting estimator.

Example 7.12 (Poisson density) Let $Y_1, \dots, Y_n$ be a Poisson random sample with mean λ , and let $\psi = \exp(-2n\lambda)$. Then an unbiased estimator $h(S)$ of ψ based on the minimal sufficient statistic $S = \sum Y_j$ must satisfy

$$\exp(-2n\lambda) = \sum_{s=0}^{\infty} h(s) \frac{(n\lambda)^s}{s!} e^{-n\lambda},$$

and completeness of S implies that the unique minimum variance unbiased estimator of ψ is the unacceptable

$$h(S) = \begin{cases} -1, & S \text{ odd,} \\ 1, & S \text{ even.} \end{cases}$$

The maximum likelihood estimator $\exp(-2S)$ is preferable despite its bias. ■

A further difficulty is that minimum variance unbiased estimators do not transform in a simple way. Moreover, as will be evident from the discussion above, there is no easy recipe that gives unbiased estimators, and once found, it may be awkward to Rao–Blackwellize them. For these and other reasons, maximum likelihood estimators are generally preferable.

7.1.4 Interval estimation

Our focus so far has been on point estimates of a parameter and their variances. Although these are useful when estimator is approximately normal, their relevance is much less obvious when its distribution is non-normal or the sample size is small. Furthermore it is often valuable to express parameter uncertainty in terms of an interval, or more generally a region. The notion of a *pivot*, which we met in Section 3.1, then moves to centre stage.

Consider a model $f(y; \theta)$ for data Y . Then a pivot $Z = z(Y, \theta)$ is a function of Y and θ that has a known distribution independent of θ , this distribution being invertible as a function of θ for each possible value of Y . That is, given a region $\mathcal{A}$ such that $\Pr\{z(Y, \theta) \in \mathcal{A}\} = 1 - 2\alpha$, we can find a region $\mathcal{R}_\alpha(Y, \mathcal{A})$ of the parameter space such that

$$1 - 2\alpha = \Pr\{z(Y, \theta) \in \mathcal{A}\} = \Pr\{\theta \in \mathcal{R}_\alpha(Y; \mathcal{A})\}.$$

If θ is scalar then $z(Y, \mathcal{A})$ is typically a strictly monotonic function of θ for each Y . Given data y and a suitable pivot, we find a $(1 - 2\alpha)$ confidence region for the true value of θ by arguing that under repeated sampling $\mathcal{R}_\alpha(y; \mathcal{A})$ is the realization of a random region $\mathcal{R}_\alpha(Y; \mathcal{A})$ that contains the true θ with probability $(1 - 2\alpha)$. An important exact pivot is the Student t statistic, and we have extensively used an approximate pivot, the likelihood ratio statistic. For reasons to be given in Section 7.3.4, pivots such as these based on the likelihood tend to be close to optimal in the sense

of providing the shortest possible confidence intervals for given α , at least in large samples.

Example 7.13 (Exponential density) Suppose we wish to base a $(1 - 2\alpha)$ confidence interval for λ on a single observation from the exponential density $\lambda e^{-\lambda y}$, $y > 0$, $\lambda > 0$. Then $Z = Y\lambda$ is pivotal, since $\Pr(\lambda Y \leq z) = 1 - e^{-z}$, $z > 0$, independent of λ . Its upper $(1 - \alpha)$ quantile is $z_{1-\alpha} = -\log \alpha$. As

$$1 - \alpha = \Pr(Z \leq z_{1-\alpha}) = \Pr(\lambda Y \leq z_{1-\alpha}) = \Pr(\lambda \leq z_{1-\alpha}/Y),$$

an upper $(1 - \alpha)$ confidence limit is $-\log \alpha/y$. Similarly an α lower confidence limit for λ is $-\log(1 - \alpha)/y$, and an equi-tailed $(1 - 2\alpha)$ confidence interval is $(-\log(1 - \alpha)/y, -\log \alpha/y)$. This is not symmetric about the maximum likelihood estimate $\hat{\lambda} = 1/y$, nor is it the shortest possible such interval.

To find the shortest $(1 - 2\alpha)$ confidence interval for λ based on y , we choose the upper tail probability γ , $0 < \gamma \leq 2\alpha$, to minimize the interval length

$$y^{-1}\{-\log \gamma + \log(1 - 2\alpha + \gamma)\},$$

giving $\gamma = 2\alpha$ and confidence interval $(0, -\log(2\alpha)/y)$. This is obvious from the shape of the exponential density and, not coincidentally, the likelihood. ■

Exercises 7.1

- 1 Let R be binomial with probability π and denominator m , and consider estimators of π of form $T = (R + a)/(m + b)$, for $a, b \geq 0$. Find a condition under which T has lower mean squared error than the maximum likelihood estimator R/m , and discuss which is preferable when $m = 5, 10$.
- 2 Let $T = a \sum (Y_j - \bar{Y})^2$ be an estimator of σ^2 based on a normal random sample. Find values of a that minimize the bias and mean squared error of T .
- 3 When T is a biased estimator of the scalar $\psi(\theta)$, with bias $b(\theta)$, show that under the usual regularity conditions, the mean squared error of T is no smaller than

$$\{d\psi/d\theta + db(\theta)/d\theta\}^2 / I(\theta) + b(\theta)^2.$$

If $b(\theta) = b_1(\theta)/n + b_2(\theta)/n^{3/2} + \dots$, where $b_i(\theta)$ is $O(1)$, then show that the Cramér–Rao lower bound applies, at least in large samples.

- 4 Suppose that T is a $q \times 1$ unbiased estimator of $\psi = \psi(\theta)$. Show that $\text{cov}(T, U) = d\psi/d\theta^T$, and compute the variance matrix of $T - d\psi/d\theta^T I(\theta)^{-1}U$, where U is $p \times 1$ score vector. Hence establish (7.3).
- 5 Consider a kernel density estimator (7.4).
 - (a) Verify the choice of h that minimizes (7.7). If $f(y) = \sigma^{-1}\phi\{(y - \mu)/\sigma\}$ and $w(u) = \phi(u)$, find h_{opt} . Discuss.
 - (b) Show that $h = 1.06\sigma n^{-1/5}$ minimises (7.8) using the densities in (a).
 - (c) Instead of using a constant bandwidth, we might take

$$\hat{f}(y) = \frac{1}{nh} \sum_{j=1}^n \frac{1}{\lambda_j} w\left(\frac{y - y_j}{h\lambda_j}\right)$$

for local bandwidth factors $\lambda_j \propto \{\tilde{f}(y_j)\}^{-\gamma}$ based on a pilot density estimate $\tilde{f}(y)$. Show that if the pilot estimate is exact and $\gamma = -\frac{1}{2}$, then $\hat{f}$ has bias $o(h^2)$.

- 6 Find the expected value of $CV(h)$, and show to what extent it estimates (7.9).

Note that
 $\phi(z)^2 = (2\pi)^{-1/2}\phi(\sqrt{2}z)$.

- 7 Find minimum variance unbiased estimators of λ^2 , e^λ , and $e^{-n\lambda}$ based on a random sample $Y_1, \dots, Y_n$ from a Poisson density with mean λ . Show that no unbiased estimator of $\log \lambda$ exists.
- 8 In Example 7.1.3, suppose we wish to estimate $\psi = \Pr(Y \leq y)$ using the empirical distribution function $n^{-1} \sum I(Y_j \leq y)$. Show that this is unbiased and that its Rao–Blackwellized form is

$$\frac{1}{n} \sum_{j=1}^n \Pr(Y_j \leq y \mid X_j).$$

Hence obtain an unbiased estimator of $f(y)$.

- 9 Let $Y \sim N(0, \theta)$. Is Y complete? What about Y^2 ? And $|Y|$?
- 10 Let $R_1, \dots, R_n$ be a binomial random sample with parameters m and $0 < \pi < 1$, where m is known. Find a complete minimal sufficient statistic for π and hence find the minimum variance unbiased estimator of $\pi(1 - \pi)$.
- 11 Let $\bar{Y}$ be the average of a random sample from the uniform density on $(0, \theta)$. Show that $2\bar{Y}$ is unbiased for θ . Find a sufficient statistic for θ , and obtain an estimator based on it which has smaller variance. Compare their mean squared errors.

7.2 Estimating Functions

7.2.1 Basic notions

Our discussion of the maximum likelihood estimator in Section 4.4.2 stressed its asymptotic properties but said little about its finite-sample behaviour. By contrast our treatment of unbiased estimators showed their finite-sample optimality under certain conditions, but suggested that the class of such estimators is often too small to be of real interest for applications. Furthermore both types of estimator can behave poorly if the data are contaminated or if the assumed model is incorrect, making it worthwhile to consider other possibilities. In this section we explore some consequences of shifting emphasis away from estimators and towards the functions that often determine them.

Suppose that we intend to estimate a $p \times 1$ parameter θ based on a random sample $Y_1, \dots, Y_n$ from a density $f(y; \theta)$, assumed to be regular for likelihood inference. Then in most cases the maximum likelihood estimator $\hat{\theta}$ is defined implicitly as the solution to the $p \times 1$ score equation

$$U(\theta) = u(Y; \theta) = \sum_{j=1}^n u(Y_j; \theta) = \sum_{j=1}^n \frac{\partial \log f(Y_j; \theta)}{\partial \theta} = 0.$$

Key properties of the score statistic $U(\theta)$ are

$$\mathbb{E}\{U(\theta)\} = 0, \quad \text{var}\{U(\theta)\} = \mathbb{E}\left\{-\frac{dU(\theta)}{d\theta^\top}\right\} = I(\theta),$$

for all θ , where the $p \times p$ Fisher information matrix $I(\theta) = ni(\theta)$ and

$$i(\theta) = \text{var}\{u(Y_j; \theta)\} = \int u(y; \theta) u(y; \theta)^\top f(y; \theta) dy = - \int \frac{\partial u(y; \theta)}{\partial \theta^\top} f(y; \theta) dy.$$

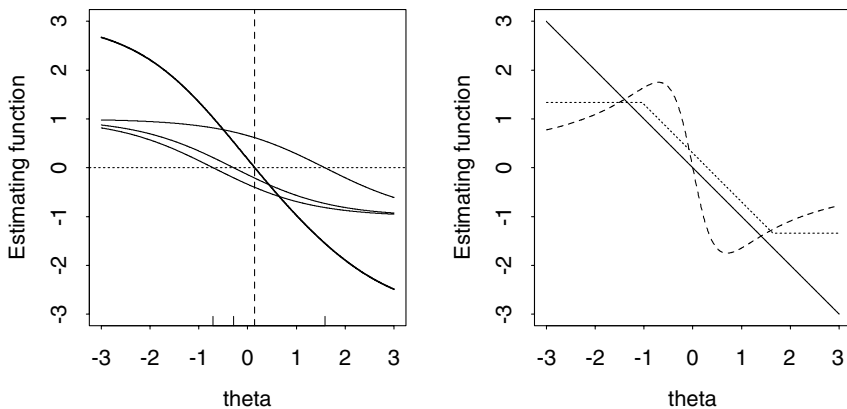


Figure 7.3 Estimating functions. Left: construction of $g(y; \theta)$ (heavy) as the sum of $g(y_j; \theta)$ for a sample of size $n = 3$ shown by the rug. The lines $g = 0$ (dots) and $\theta = \tilde{\theta}$ (dashes) are also shown. Right: estimating functions for the mean (solid), the Huber estimator (dots) and a redescending M -estimator (dashes), slightly offset to avoid overplotting.

The implicit definition of $\hat{\theta}$ suggests that we study properties of estimators $\tilde{\theta}$ that solve a $p \times 1$ system of *estimating equations* of form

$$g(Y; \theta) = \sum_{j=1}^n g(Y_j; \theta) = 0. \quad (7.15)$$

We call $g(y; \theta)$ an *estimating function* and say it is unbiased if

$$E\{g(Y; \theta)\} = n \int g(y; \theta) f(y; \theta) dy = 0 \quad \text{for all } \theta.$$

Or sometimes an inference function.

This formulation encompasses many possibilities.

Example 7.14 (Logistic density) The logistic density $e^{y-\theta}/(1+e^{y-\theta})^2$ has score function

$$u(y; \theta) = 2e^{y-\theta}/(1+e^{y-\theta}) - 1, \quad -\infty < y < \infty, -\infty < \theta < \infty.$$

The left panel of Figure 7.3 shows the construction of the corresponding estimating function based on a sample of size three. ■

Example 7.15 (Moment estimators) If $g(y; \mu) = y - \mu$, then the solution to (7.15) is the sample average $\tilde{\mu} = \bar{Y}$, which is an unbiased estimator of the mean of f , if this exists. The estimating function $y - \mu$ is shown in the right panel of Figure 7.3, with other estimating functions discussed later.

This can be extended to several parameters. The *moment estimators* of the mean and variance of Y are found by simultaneous solution of

Or method of moments estimators.

$$n^{-1} \sum_{j=1}^n Y_j - \mu = 0, \quad n^{-1} \sum_{j=1}^n Y_j^2 - \mu^2 - \sigma^2 = 0,$$

and these are of form (7.15) with $g(y; \theta) = (y - \mu, y^2 - \mu^2 - \sigma^2)^T$ and $\theta = (\mu, \sigma^2)^T$. Although themselves unbiased, these estimating equations produce the biased estimator $n^{-1} \sum (Y_j - \bar{Y})^2$ of σ^2 .

Estimators of functions of the mean and variance may be defined similarly. For example, the Weibull density

$$f(y; \beta, \kappa) = \kappa \beta^{-1} (y/\beta)^{\kappa-1} \exp\{-(y/\beta)^\kappa\}, \quad y > 0, \beta, \kappa > 0,$$

$\Gamma(u)$ is the gamma function.

has $E(Y^r) = \beta^r \Gamma(1 + r/\kappa)$. Hence the moment estimator of $\theta = (\beta, \kappa)^T$ can be determined as the solution to (7.15) with

$$g(y; \theta) = (y - \beta \Gamma(1 + 1/\kappa), \quad y^2 - \beta^2 \Gamma(1 + 2/\kappa))^T. \quad (7.16)$$

The parameters μ and σ^2 have the same interpretations for any model that possesses two moments, whereas (β, κ) are specific to the Weibull case. ■

Example 7.16 (Probability weighted moment estimators) Moment estimators may be poor or even useless with data from long-tailed densities, whose moments may not exist. An alternative is use of *probability weighted moment estimators*, defined as solutions to equations of form

$$n^{-1} \sum_{j=1}^n Y_j^r F(Y_j; \theta)^s \{1 - F(Y_j; \theta)\}^t - \int y^r F(y; \theta)^s \{1 - F(y; \theta)\}^t f(y; \theta) dy = 0.$$

Even if the ordinary moments, which correspond to taking $s = t = 0$, do not exist, the integrals here may be finite for positive values of s or t or both.

An example is the generalized Pareto distribution (6.38), for which we set $\theta = (\xi, \sigma)^T$. In this case it is convenient to take $r = 1$ and $s = 0$, giving

$$g_t(y; \theta) = y(1 + \xi y/\sigma)^{-t/\xi} - \frac{\sigma}{(t+1)(t+1-\xi)},$$

which has finite expectation provided $\xi < t + 1$. Estimators may be obtained by setting $g(y; \theta) = (g_1(y; \theta), g_2(y; \theta))^T$ and solving (7.15) simultaneously, though equivalent more convenient forms of the equations are preferred in practice.

As with moment estimators, the choice of r , s , and t introduces an arbitrary element, because different choices will lead to different estimators. ■

Example 7.17 (Linear model) The scalar β in the simple linear model

$$Y_j = \beta x_j + \varepsilon_j, \quad j = 1, \dots, n,$$

where the ε_j have mean zero, can be estimated by the solution to (7.15) with $g(y; \theta) = y - \beta x$, giving $\tilde{\beta} = \sum Y_j / \sum x_j$. This estimator is unbiased whatever the distributions of the ε_j ; in particular we have made no assumptions about their variances, requiring the ε_j only to have zero mean. In fact, they need not be independent, or even uncorrelated. ■

In general discussion we shall suppose that θ is scalar and that for every value of y , we deal with an unbiased estimating function $g(y; \theta)$ that is strictly monotone decreasing in θ . It is then easy to show that $\tilde{\theta}$ is consistent for θ . Note first that $\tilde{\theta} \leq a$ if and only if $g(Y; a) \leq 0$. As $g(y; \theta)$ is decreasing in θ for each y , $n^{-1} g(Y; \theta - \varepsilon)$

converges to

$$n^{-1}E\{g(Y; \theta - \varepsilon)\} = n^{-1}E\{g(Y; \theta - \varepsilon) - g(Y; \theta)\} = c(\theta - \varepsilon) > 0$$

as $n \rightarrow \infty$ for any $\varepsilon > 0$, by virtue of the weak law of large numbers. Hence

$$\Pr(\tilde{\theta} \leq \theta - \varepsilon) = \Pr\{n^{-1}g(Y; \theta - \varepsilon) \leq 0\} \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

Likewise $\Pr(\tilde{\theta} > \theta + \varepsilon) \rightarrow 0$, so $\Pr(|\tilde{\theta} - \theta| \leq \varepsilon) \rightarrow 1$: $\tilde{\theta}$ is a consistent estimator.

Technical difficulties arise with non-monotone or discontinuous estimating functions, to which most of the discussion below does not apply directly. In such cases it is necessary to show that there is a consistent solution to the estimating equation, to which the arguments below can be applied.

Optimality

Having defined the class of unbiased estimating functions, the question naturally arises which of them we should use. To answer this we must find a finite-sample optimality criterion analogous to mean squared error. To motivate a suitable criterion, suppose that θ is scalar and consider its estimator $\tilde{\theta}$. Taylor series expansion of $g(Y; \tilde{\theta})$ gives

$$0 \doteq g(Y; \theta) + (\tilde{\theta} - \theta) \frac{dg(Y; \theta)}{d\theta},$$

so

$$\tilde{\theta} - \theta \doteq \frac{\sum_{j=1}^n g(Y_j; \theta)}{-\sum_{j=1}^n \frac{dg(Y_j; \theta)}{d\theta}} = \frac{\sum_{j=1}^n g(Y_j; \theta)}{E\left\{-\frac{dg(Y; \theta)}{d\theta}\right\}} + O_p(n^{-1}), \quad (7.17)$$

using the same argument as applied to the maximum likelihood estimator. This implies that $\tilde{\theta}$ has asymptotic variance

$$\text{var}(\tilde{\theta}) \doteq \frac{\text{var}\{g(Y; \theta)\}}{\left[E\left\{-\frac{dg(Y; \theta)}{d\theta}\right\}\right]^2} = n^{-1} \frac{\int g(y; \theta)^2 f(y; \theta) dy}{\left\{-\int \frac{dg(y; \theta)}{d\theta} f(y; \theta) dy\right\}^2}.$$

A measure of finite-sample performance of $g(y; \theta)$ should not conflict with asymptotic properties of $\tilde{\theta}$, suggesting that we regard an estimating function as optimal in the class of unbiased estimating functions if it minimizes

$$\frac{\text{var}\{g(Y; \theta)\}}{\left[E\left\{-\frac{dg(Y; \theta)}{d\theta}\right\}\right]^2} \quad (7.18)$$

for all θ . This quantity is unaffected by one-one reparametrization.

Another motivation for (7.18) rests on noting that although variance is a natural basis for comparing estimating functions, $ag(Y; \theta)$ is also unbiased, with variance a^2 times greater than that of $g(Y; \theta)$. Hence fair comparison is possible only after removing this arbitrary scaling. Multiplication of $g(Y; \theta)$ by a changes the slope of the estimating function, so it is natural to choose a to ensure that the expected derivative of $g(Y; \theta)$ equals one, leading to (7.18).

It can be shown that any unbiased estimating function must satisfy

$$I(\theta)^{-1} \leq \frac{\text{var}\{g(Y; \theta)\}}{\left[E\left\{-\frac{dg(Y; \theta)}{d\theta}\right\}\right]^2}, \quad (7.19)$$

so there is a lower bound on (7.18), analogous to the Cramér–Rao lower bound. If (7.18) is evaluated with $g(Y; \theta) = u(Y; \theta)$, the result is $I(\theta)^{-1}$. Hence the score function minimizes (7.18), and is in this sense optimal in finite samples. This ties in with asymptotic properties of the maximum likelihood estimator, and may be extended to the case where θ is a $p \times 1$ vector. Then

$$E\left\{-\frac{\partial g(Y; \theta)}{\partial \theta^T}\right\}^{-1} \text{var}\{g(Y; \theta)\} E\left\{-\frac{\partial g(Y; \theta)^T}{\partial \theta}\right\}^{-1} \geq I(\theta)^{-1} \quad (7.20)$$

in the sense that the difference of these $p \times p$ matrices is positive semi-definite, provided $E\{-\partial g(Y; \theta)/\partial \theta^T\}$ is invertible. The left-hand side of this inequality is the asymptotic covariance matrix of $\tilde{\theta}$, and its sandwich form generalizes that of a maximum likelihood estimator under a wrong model; see Section 4.6. Standard errors for $\tilde{\theta}$ are obtained by replacing the matrices in (7.20) by sample versions, giving

$$\left\{\sum_{j=1}^n \frac{\partial g(y_j; \tilde{\theta})}{\partial \theta^T}\right\}^{-1} \sum_{j=1}^n g(y_j; \tilde{\theta}) g(y_j; \tilde{\theta})^T \left\{\sum_{j=1}^n \frac{\partial g(y_j; \tilde{\theta})}{\partial \theta}\right\}^{-1},$$

from which confidence sets for elements of θ may be obtained, generally by normal approximation.

Example 7.18 (Weibull model) An estimating function for the Weibull parameters β and κ is given by (7.16), for which elementary calculations give

$$E\left\{-\frac{\partial g(Y; \theta)^T}{\partial \theta}\right\} = n \begin{pmatrix} \Gamma(1 + 1/\kappa) & 2\beta\Gamma(1 + 2/\kappa) \\ -\beta\Gamma'(1 + 1/\kappa)/\kappa^2 & -2\beta^2\Gamma'(1 + 2/\kappa)/\kappa^2 \end{pmatrix}$$

$\Gamma'(u) = d\Gamma(u)/du$, and so forth.

and

$$I(\theta) = n \begin{pmatrix} \kappa^2/\beta^2 & -\Gamma'(2)/\beta \\ -\Gamma'(2)/\beta & \{1 + \Gamma''(2)\}/\kappa^2 \end{pmatrix},$$

while $\text{var}\{g(Y; \theta)\}$ is easily found in terms of the moments $E(Y^r)$. In analogy to the discussion of efficiency on page 113, the overall efficiency of $g(Y; \theta)$ relative to the score is taken to be the square root of the ratio of the determinants of the matrices on either side of the inequality in (7.20), while the efficiency for estimation of β is the ratio of their (1, 1) coefficients, with (2, 2) coefficients used for κ . These efficiencies, plotted in the left panel of Figure 7.4, show that the moment estimating functions are fairly efficient when $\kappa > 2$, but are poor when κ is small. ■

7.2.2 Robustness

Finite-sample optimality of the score function is not the whole story, for several reasons. First, we may be unwilling or unable to specify the model fully, and then the score is unavailable. Second, even if we can be fairly sure of $f(y; \theta)$, there is

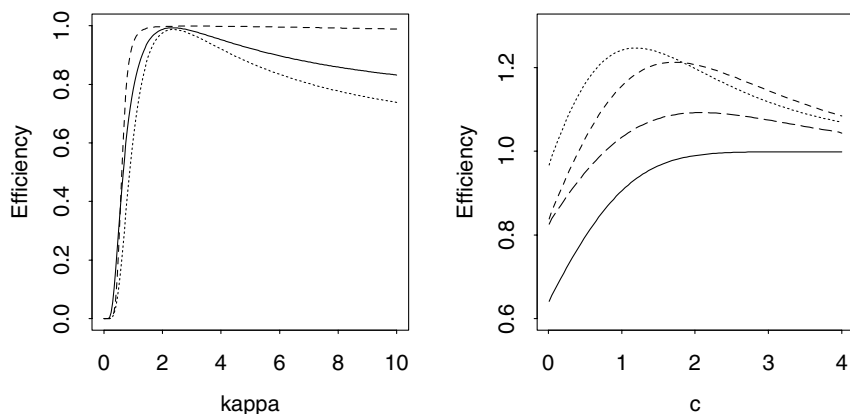


Figure 7.4 Efficiencies of estimating functions. Left: overall efficiency (solid), efficiency for β (dashes) and for κ (dots) for moment estimators of Weibull distribution. Right: finite-sample efficiency of Huber estimating function g_c relative to $g(y; \theta) = y - \theta$ for normal (solid), t_5 (dots), normal mixture $0.95N(0, 1) + 0.05N(0, 9)$ (small dashes) and logistic data (long dashes).

always the possibility of bad data — trying errors, wild observations and so forth. In principle all data should be carefully scrutinized for these, but with big or complex datasets or where data are collected automatically this is impracticable. Estimating functions that are *robust*, that is, perform well under a wide range of potential models centred at an ideal model may be preferred, even if they are somewhat sub-optimal when that model itself holds.

Robustness entails insensitivity to departures from assumptions, but this has many aspects. Perhaps the most common usage relates to contamination by outliers. If bad values are present then we might optimistically hope to identify and delete them, or more realistically aim to downweight them. Thus we ignore or play down some ‘bad’ portion of the data and hope to extract useful information from the ‘good’ part, even if we are unsure where the boundary lies. A related usage concerns the need for procedures that perform well when assumptions underlying the ideal model are relaxed. An essential requirement is then that estimands have the same interpretation under all the potential models. In Example 7.15 the first and second moments μ and σ^2 have this property of *robustness of interpretation* but the Weibull parameters κ and β do not, because they are meaningless for models other than the Weibull.

Outliers are perhaps the most obvious form of departure from the model, but the assumed dependence structure is usually more crucial in applications. In Example 6.25, for instance, a confidence interval was three times too short when dependence was unaccounted for. Although independence is often assumed, not only is mild dependence often difficult to detect, but also it may be hard to formulate a suitable alternative. In applications independence may be assured by the design of the investigation, but often it must be checked empirically, for example using time series tools such as the correlogram.

One way to view an estimating function is that it defines a parameter $t(F)$ implicitly as the solution to the population equation

$$\int g\{y; t(F)\} dF(y) = 0,$$

where F is any member of the class of distributions under consideration. The requirement that $t(F)$ be robust of interpretation imposes restrictions on g . If, for instance, the density $f(y) = dF(y)/dy$ is symmetric about θ and we require $t(F) = \theta$ for any such density, then $g(y; \theta)$ must be odd as a function of $y - \theta$, with $g(\theta; \theta) = 0$. In many cases the requirement of robustness of interpretation indicates taking $t(F)$ to be a moment or related quantity, which will retain its meaning for all models possessing the necessary moments.

One approach to downweighting bad data stems from observing that (7.17) implies that the effect of Y_j on $\tilde{\theta}$ is proportional to $g(Y_j; \theta)$. If this is large, then $\tilde{\theta}$ will tend to be far from its estimand θ . This suggests that the sensitivity of $\tilde{\theta}$ to an observation y be measured by the *influence function* of $\tilde{\theta}$,

$$L(y; \theta) = \frac{g(y; \theta)}{-\int \frac{dg(u; \theta)}{d\theta} f(u; \theta) du};$$

this is simply a rescaling of the estimating function. Our earlier discussion implies that $\text{var}(\tilde{\theta}) \doteq n^{-1} \text{var}\{L(Y; \theta)\}$ in terms of a single observation Y .

Expression (7.17) suggests that the impact of outliers can be reduced by using estimating functions and hence influence functions that are bounded in y . One possibility is a redescending function such as $(y - \theta)/\{1 + (y - \theta)^2\}$, which tends to zero as $|y - \theta| \rightarrow \infty$. Another possibility is to truncate a standard function such as $y - \theta$, so that values of y distant from θ have limited impact on $\tilde{\theta}$. See Figure 7.3.

Example 7.19 (Huber estimator) The effect of outliers on the estimation of a mean may be reduced by using

$$g_c(y; \theta) = \begin{cases} -c, & y \leq \theta - c, \\ y - \theta, & -c < y - \theta < c, \\ c, & \theta + c \leq y, \end{cases}$$

where the constant $c > 0$ is chosen to balance robustness and efficiency. Robustness to outliers is increased but efficiency at the normal model is reduced by decreasing c ; when $c = \infty$ we have $g_\infty(y; \theta) = y - \theta$ and $\tilde{\theta} = \bar{Y}$. The estimator corresponding to $g_c(y; \theta)$ is sometimes called the *Huber estimator* of location. The parameter $t(F)$ is the centre of an underlying symmetric density and equals its mean when $c = \infty$ and its median when $c = 0$. These are not the same when the underlying density is asymmetric, and then $t(F)$ has no simple direct interpretation, though it may depend only weakly on c for certain choices of F .

The finite-sample efficiency of $g_c(y; \theta)$ as a function of c for various symmetric densities is shown in the right panel of Figure 7.4. The quantity plotted is (7.18) divided by the variance of $g_\infty(Y; \theta) = Y - \theta$, as this rather than the score function for the true density would usually be used in practice. Under the normal model the efficiency of g_c is essentially one when $c = 2$, dropping to the value $2/\pi = 0.637$ for the median when $c \rightarrow 0$. Overall a good choice seems to be $c = 1.345$, which is often the default in software packages; it has efficiency 0.95 for normal data, but beats g_∞ in the other cases shown. ■

Peter Johann Huber (1934–) has been professor of statistics at ETH Zürich, Massachusetts Institute of Technology, and Harvard and Bayreuth universities, and is now retired.

Or Huber's Proposal 2.

The discussion above presupposes that the scale of the underlying density is known, even if the location is not. In practice estimation of scale has little effect on the efficiency of location estimators, and the results above apply with little change provided scale is estimated robustly, for example using the median absolute deviation.

To illustrate optimality under weak conditions on the underlying model, suppose that we intend to estimate θ using the weighted combination of unbiased linear estimating functions

$$\sum_{j=1}^m w_j(\theta) \{Y_j - \mu_j(\theta)\},$$

where $\text{var}(Y_j) = V_j(\theta)$ may be a function of θ . We suppose that the mean and variance functions $\mu_j(\theta)$ and $V_j(\theta)$ for each of the Y_j are known, but make no assumption about their distributions. Notice that our argument for consistency of $\tilde{\theta}$ will apply under mild conditions on the weights and the moments. Suppose also that the Y_j are uncorrelated. Then (7.18) is

$$\frac{\sum_j w_j^2(\theta) V_j(\theta)}{\{\sum_j w_j(\theta) \mu'_j(\theta)\}^2},$$

where $\mu'_j(\theta) = d\mu_j(\theta)/d\theta$, and our earlier discussion suggests that we seek the weights $w_j(\theta)$ that minimize this. This is equivalent to the problem

$$\min_{w_1, \dots, w_n} \sum_{j=1}^n w_j^2 V_j \quad \text{subject to} \quad \sum_{j=1}^n w_j \mu'_j = c,$$

for some constant c . Use of Lagrange multipliers gives $w_j(\theta) \propto \mu'_j(\theta)/V_j(\theta)$, so the optimal estimating equation is

$$\sum_{j=1}^n \mu'_j(\theta) \frac{1}{V_j(\theta)} \{Y_j - \mu_j(\theta)\} = 0. \quad (7.21)$$

An exponential family variable Y_j with log likelihood contribution $y_j\theta - \kappa_j(\theta)$ has mean $\kappa'_j(\theta)$ and variance $\kappa''_j(\theta)$, so $\mu'_j(\theta) = V_j(\theta)$ and (7.21) reduces to the score equation, $\sum \{Y_j - \kappa'_j(\theta)\} = 0$, which is optimal.

Example 7.20 (Straight-line regression) Let the Y_j have means $\mu(\beta) = x_j\beta$, with x_j known. Then $\mu'_j(\beta) = x_j$, and $g(Y_j, \beta) = Y_j - x_j\beta$. If $\text{var}(Y_j) = V_j(\beta)$ is constant, (7.21) becomes $\sum x_j(Y_j - \beta x_j)$, and the corresponding estimator is $\tilde{\beta} = \sum Y_j x_j / \sum x_j^2$. This is the least squares estimator of β , corresponding to a normal distribution for Y_j , but it has much wider validity.

If $\text{var}(Y_j) = x_j\beta$, as would be the case if Y_j were Poisson with mean $x_j\beta$, then the optimal estimating function is $\sum(Y_j - \beta x_j)$, and $\tilde{\beta} = \sum Y_j / \sum x_j$. As in the normal case, $\tilde{\beta}$ is optimal more widely. ■

Estimating equations of form similar to (7.21) are very important in the regression models encountered in Chapters 8 and 10.

This may be omitted at a first reading.

7.2.3 Dependent data

In earlier discussion, for example in Section 6.1, we used the fact that standard likelihood asymptotics also apply to some types of dependent data. For some explanation of this, consider the more general context of unbiased estimating functions for a scalar θ . Suppose that $\tilde{\theta}$ is defined as the solution to the equation

$$\sum_{j=1}^n g_j(Y; \theta) = 0, \quad (7.22)$$

where $g_j(Y; \theta)$ depends only on $Y_1, \dots, Y_j$ and is such that for all θ ,

$$E\{g_1(Y)\} = 0, \quad E\{g_j(Y; \theta) \mid Y_1, \dots, Y_{j-1}\} = 0, \quad j = 2, \dots, n,$$

so that the unconditional expectation $E\{g_j(Y; \theta)\} = 0$ for all j . If $j > k$, then

$$\begin{aligned} \text{cov}\{g_j(Y; \theta), g_k(Y; \theta)\} &= E\{g_j(Y; \theta)g_k(Y; \theta)\} \\ &= E\{g_k(Y; \theta)E\{g_j(Y; \theta) \mid Y_1, \dots, Y_{j-1}\}\} = 0, \end{aligned}$$

so

$$\text{var} \left\{ \sum_{j=1}^n g_j(Y; \theta) \right\} = \sum_{j=1}^n \text{var}\{g_j(Y; \theta)\}.$$

The left of (7.22) is a zero-mean martingale, and under mild regularity conditions a martingale central limit theorem as $n \rightarrow \infty$ gives

$$V^{-1/2}(\tilde{\theta} - \theta) \xrightarrow{D} Z, \quad \text{where } V = \frac{\sum_{j=1}^n \text{var}\{g_j(Y; \theta) \mid Y_1, \dots, Y_{j-1}\}}{\left[\sum_{j=1}^n E\{dg_j(Y; \theta)/d\theta \mid Y_1, \dots, Y_{j-1}\} \right]^2}, \quad (7.23)$$

and Z is standard normal. Thus provided the random variable V is used to estimate the variance of $\tilde{\theta}$, confidence intervals for θ can be set in the usual way.

Two main possibilities arise for the limiting behaviour of V . In an *ergodic* model a deterministically rescaled version of V converges to a constant as $n \rightarrow \infty$, such as $nV \xrightarrow{P} \nu > 0$. This occurs, for instance, with independent data, ergodic Markov chains, and many time series models. Under regularity conditions the usual arguments then apply to the rescaled estimator, whose limiting distribution is normal, and the argument starting from (7.17) yields (7.18). The second possibility is that when rescaled, V converges to a nondegenerate random variable D . The model is then said to be *non-ergodic*, and as the limiting distribution of the rescaled estimator is $D^{-1/2}Z$, standard large-sample theory does not apply.

As with independent data, we can find the optimal finite-sample choice of weighting functions within the class of linear combinations of the $g_j(Y; \theta)$,

$$\sum_{j=1}^n W_j(\theta) g_j(Y; \theta),$$

where the $W_j(\theta)$, now random variables, can depend on $Y_1, \dots, Y_{j-1}$ and θ . This

turns out to be

$$W_j(\theta) = \frac{-E\{dg_j(Y; \theta)/d\theta \mid Y_1, \dots, Y_{j-1}\}}{\text{var}\{g_j(Y; \theta) \mid Y_1, \dots, Y_{j-1}\}}. \quad (7.24)$$

This finite-sample result is independent of the asymptotic properties of $\tilde{\theta}$.

Example 7.21 (Branching process) The branching process was first used to model the survival of surnames, it being supposed that a surname would die out if all every male bearing it had no sons, but it has applications in epidemic modelling and elsewhere. Each of the Y_{j-1} individuals in generation $j - 1$ independently gives birth to a random number of individuals, so $Y_j = \sum_{i=1}^{Y_{j-1}} N_i$, where the N_i are independent with mean θ and variance σ^2 . We take $Y_0 = 1$. Here $g_j(Y; \theta) = Y_j - \theta Y_{j-1}$ is unbiased whatever the distribution of the N_i , while

$$\text{var}\{g_j(Y; \theta) \mid Y_1, \dots, Y_{j-1}\} = Y_{j-1}\sigma^2, \quad E\left\{-\frac{dg_j(Y; \theta)}{d\theta} \mid Y_1, \dots, Y_{j-1}\right\} = Y_{j-1}.$$

The optimal weights are $W_j(\theta) = 1/\sigma^2$, here non-random, and the corresponding estimating equation is $\sum_{j=2}^n (Y_j - \theta Y_{j-1}) = 0$, whatever the distribution of the N_i . Thus $\tilde{\theta} = \sum_{j=1}^{n-1} Y_{j+1} / \sum_{j=1}^{n-1} Y_j$ is optimal and $V = \sigma^2 / \sum_{j=1}^n Y_{j-1}$.

Extinction is certain if $\theta \leq 1$ but not if $\theta > 1$. If extinction occurs then no estimator of θ can be consistent. When $\theta > 1$ and given that extinction does not occur, (7.23) implies that $V^{-1/2}(\tilde{\theta} - \theta) \xrightarrow{D} \sigma Z$. In this case $\theta^{-n}V$ converges to a nondegenerate random variable and the asymptotics are nonstandard. Confidence intervals for θ are best constructed using V .

Other growth models such as birth processes and non-stationary diffusions can also be non-ergodic. As the discussion above suggests, inference for θ is then best performed using observed information or its generalization V^{-1} . ■

The argument leading to (7.23) applies in particular to maximum likelihood estimators. We write $f(y_1, \dots, y_n; \theta) = f(y_1; \theta) \prod_{j=2}^n f(y_j \mid y_1, \dots, y_{j-1}; \theta)$ and express the score as

$$\frac{d\ell(\theta)}{d\theta} = \frac{d \log f(Y_1; \theta)}{d\theta} + \sum_{j=2}^n \frac{d \log f(Y_j \mid Y_1, \dots, Y_{j-1}; \theta)}{d\theta} = \sum_{j=1}^n g_j(Y; \theta).$$

Here $W_j(\theta) \equiv 1$, so the unweighted score is optimal in finite samples. In the ergodic case, Taylor series arguments establish the usual properties of maximum likelihood estimators and likelihood ratio statistics, subject to regularity conditions like those needed for independent data.

Exercises 7.2

- 1 Show that if an estimating function undergoes a smooth 1–1 reparametrization by writing $g(y; \theta) = g(y; \theta(\psi)) = g'(y; \psi)$, then $\tilde{\theta} = \theta(\tilde{\psi})$. Establish also that (7.18) is unchanged.
- 2 Show that the sample median of a continuous density solves (7.15) with

$$g(y; \theta) = H(y - \theta) - H(\theta - y),$$

$H(u)$ is the Heaviside function.

giving $g(Y; \theta) = \sum \{I(\theta \leq Y_j) - I(Y_j \leq \theta)\}$, a descending staircase, with a unique solution only when n is odd. Find (7.18). Surprised?

- 3 Find the form of estimating function for an exponential family model.
- 4 To verify (7.17), show that the numerator and denominator in the first ratio may be written as $n^{1/2}\varepsilon_n$ and $n\zeta + n^{1/2}\eta_n$, where $\zeta \neq 0$ and ε_n and η_n are $O_p(1)$ random variables. Deduce that the ratio is $n^{-1/2}\varepsilon_n\zeta^{-1}(1 - n^{-1/2}\eta_n\zeta^{-1} + \dots)$, and hence find the desired result.
- 5 Reread the proof of the Cramér–Rao lower bound, and then establish (7.19).
- 6 To establish (7.20), let C and G denote the $p \times p$ matrix $E\{-\partial g(Y; \theta)^T / \partial \theta\}$ and the $p \times 1$ vector $g(Y; \theta)$, note that $C = \text{cov}\{G, U(\theta)\}$ and, assuming that C is invertible, compute the variance matrix of $C^{-1}G - I(\theta)^{-1}U(\theta)$.
- 7 Let F_ν represent the gamma distribution with unit mean and shape parameter ν . Investigate how the quantity $t(F_\nu)$ determined by the Huber estimating function $g_c(y; \theta)$ depends on c and ν .
- 8 To establish (7.24), note that (7.18) depends on

$$E \left\{ \sum_{j=1}^n w_j^2 E_{j-1} (G_j^2) \right\}, \quad E \left\{ \sum_{j=1}^n w_j^2 E_{j-1} \left(\frac{dG_j}{d\theta} \right) \right\},$$

where E_{j-1} denotes expectation conditional on $Y_1, \dots, Y_{j-1}$ and $G_j = g_j(Y; \theta)$. Call the sums here A^2 and B , so that (7.18) has inverse $\{E(B)\}^2/E(A^2)$.

- (a) Use the fact that $E\{(B/A - cA)^2\} \geq 0$ to show that $E(B)^2/E(A^2) \leq E(B^2/A^2)$.
- (b) Deduce that $E(B^2/A^2)$ is maximized by (7.24), and show that this choice gives $E(B)^2/E(A^2) = E(B^2/A^2)$.
- (c) Hence show that (7.18) is minimized among the class of estimating functions $\sum w_j(\theta)g_j(Y; \theta)$ by taking (7.24). (Godambe, 1985)
- 9 Find the optimal estimating function based on dependent data $Y_1, \dots, Y_n$ with $g_j(Y; \theta) = Y_j - \theta Y_{j-1}$ and $\text{var}\{g_j(Y; \theta) \mid Y_1, \dots, Y_{j-1}\} = \sigma^2$. Derive also the estimator $\hat{\theta}$. Find the maximum likelihood estimator of θ when the conditional density of Y_j given the past is $N(\theta y_{j-1}, \sigma^2)$. Discuss.

7.3 Hypothesis Tests

7.3.1 Significance levels

A scientific theory or hypothesis leads to assertions that are testable using empirical data. Such data may discredit the hypothesis, as when the Michelson–Morley experiment demolished the nineteenth-century notion of an aether in which the earth and planets move, or they may lead to elaboration or development of it, just as quantum theory supercedes Newtonian mechanics but does not make Newton’s laws of motion useless for daily life. One way to investigate the extent to which an assertion is supported by the data Y is to choose a *test statistic*, $T = t(Y)$, large values of which cast doubt on the assertion and hence on the underlying theory. This theory, the *null hypothesis* H_0 , places restrictions on the distribution of Y and is used to calculate a *significance level* or *P-value*

$$p_{\text{obs}} = \text{Pr}_0(T \geq t_{\text{obs}}), \quad (7.25)$$

where t_{obs} is the value of T actually observed. A distribution computed under the assumption that H_0 is true is called a *null distribution*, and then we use $\Pr_0, E_0, \dots$ to indicate probability, expectation and so forth. Small values of p_{obs} correspond to values t_{obs} unlikely to arise under H_0 , and signal that theory and data are inconsistent. The rationale for calculating the probability that $T \geq t_{\text{obs}}$ in (7.25) is that any value $t' > t_{\text{obs}}$ would cast even greater doubt on H_0 . A hypothesis that completely determines the distribution of Y is called *simple*; otherwise it is *composite*.

If there is a precise idea what situation will hold if the null hypothesis is false, then there is a clearly specified *alternative hypothesis*, H_1 , and we can choose a test statistic that has high probability of detecting departures from H_0 in the direction of H_1 . Otherwise the alternative may be very vague. In either case calculation of (7.25) involves only H_0 .

For many standard tests the null distribution of T is tabulated, available in statistical packages, or readily approximated. If not, (7.25) can be estimated by generating R independent sets of data Y_r^* from the null distribution of Y , calculating the corresponding values $T_r^* = t(Y_r^*)$, and then setting

$$\hat{p}_{\text{obs}} = \frac{1 + \sum_{r=1}^R I(T_r^* \geq t_{\text{obs}})}{1 + R}; \quad (7.26)$$

the added 1s here arise because under H_0 the original value t_{obs} is a realization of T and trivially $t_{\text{obs}} \geq t_{\text{obs}}$. The indicators $I(T_r^* \geq t_{\text{obs}})$ are independent Bernoulli variables with probability p_{obs} under H_0 , and this enables a suitable R to be determined (Exercise 7.3.1).

Example 7.22 (Exponential density) Consider an exponential random sample $Y_1, \dots, Y_n$ with parameter λ . We wish to test $\lambda = \lambda_0$ against the alternative $\lambda = \lambda_1$, with both λ_0 and λ_1 known, using the likelihood ratio

$$T' = \frac{\lambda_1^n \exp(-\lambda_1 \sum Y_j)}{\lambda_0^n \exp(-\lambda_0 \sum Y_j)} = \exp \left\{ (\lambda_0 - \lambda_1) \sum_{j=1}^n Y_j + n \log(\lambda_1/\lambda_0) \right\}.$$

We declare that doubt is cast on λ_0 if T' or equivalently $(\lambda_0 - \lambda_1) \sum Y_j$ is large. If $\lambda_1 < \lambda_0$, the value of p_{obs} is $\Pr_0(\sum Y_j > t_{\text{obs}})$, where $t_{\text{obs}} = \sum y_j$. Under the null hypothesis, $\sum Y_j$ has a gamma distribution with index n and rate λ_0 , so if $\lambda_1 < \lambda_0$, the P-value is

$$p_{\text{obs}} = \int_{t_{\text{obs}}}^{\infty} \frac{\lambda_0^n u^{n-1}}{\Gamma(n)} e^{-\lambda_0 u} du = \int_{\lambda_0 t_{\text{obs}}}^{\infty} \frac{v^{n-1}}{\Gamma(n)} e^{-v} dv = \Pr(V \geq \lambda_0 t_{\text{obs}}),$$

where V has a gamma distribution with index n ; p_{obs} can be calculated exactly because λ_0 and t_{obs} are known. ■

Examples of situations with a vague alternative hypothesis are given below.

Interpretation

The significance level may be written as $p_{\text{obs}} = 1 - F_0(t_{\text{obs}})$, where F_0 is the null distribution function of T , supposed to be continuous. One interpretation of p_{obs}

stems from the corresponding random variable, $P = 1 - F_0(T)$. For $0 \leq u \leq 1$, its null distribution is

$$\begin{aligned}\Pr_0 \{1 - F_0(T) \leq u\} &= \Pr_0 \{F_0^{-1}(1 - u) \leq T\} \\ &= 1 - F_0 \{F_0^{-1}(1 - u)\} = u,\end{aligned}$$

that is, uniform on the unit interval. Hence if we regard the observed t_{obs} as being just decisive evidence against H_0 , then this is equivalent to following a procedure which rejects H_0 with error rate p_{obs} : if we tested many different hypotheses and rejected them all, the same t_{obs} having arisen in each case, then a proportion p_{obs} of our decisions would be incorrect. This interpretation applies exactly if F_0 is known, and the test is then called *exact*; otherwise it will typically apply only as an approximation in large samples.

A common misinterpretation of the P-value is as the probability that the null hypothesis is true. This cannot be the case, because alternative hypotheses play no direct role in its calculation. Bayesian P-values account for alternatives and do have this more direct interpretation; see Section 11.2.2.

Hypothesis testing is very useful in certain contexts but has important limitations. A first is that statistical significance of a result may be quite different from its practical importance, because even a very small p_{obs} may correspond to an uninteresting departure from the null hypothesis. For example, a test for lack of fit of a parametric model may be highly significant even though the model is satisfactory, simply because the fit is poor only in an unimportant part of the distribution or because the sample size is so large that no simple parametric model can be expected to fit well. On the other hand a large value of p_{obs} may arise when effects of real importance are undetectable because the sample size is too small. Computer models of climate change suggest that rare weather events may be occurring more frequently, for example, but most daily temperature series are too short to detect such small changes.

A second limitation is that even a very small P-value may sometimes indicate more support for the null than for an alternative hypothesis. A simple test of the null hypothesis $\mu = 0$ based on a single $N(\mu, 1)$ random variable with value $y = 3$ against the alternative hypothesis $\mu = 20$ has significance level $1 - \Phi(y) \doteq 0.001$, but $\mu = 0$ is clearly more plausible than $\mu = 20$.

A third limitation is that a P-value simply gives evidence against the null hypothesis and does not indicate which of a family of alternatives is best supported by the data. For this reason the use of confidence intervals for model parameters is generally preferable, when it is feasible.

Goodness of fit tests

In earlier chapters we used graphs such as probability plots to assess model fit. We now briefly discuss how to supplement such informal procedures with more formal ones. Suppose initially that the null hypothesis is that a random sample $Y_1, \dots, Y_n$ has issued from a known continuous distribution $F(y)$. Then we can compare F with

the empirical distribution function

$$\widehat{F}(y) = n^{-1} \sum_{j=1}^n I(Y_j \leq y),$$

whose mean and variance are $F(y)$ and $F(y)\{1 - F(y)\}/n$ under H_0 .

Standard measures of distance between F and $\widehat{F}$ include the Kolmogorov–Smirnov, Cramér–von Mises and Anderson–Darling statistics

$$\begin{aligned} \sup_y |\widehat{F}(y) - F(y)| &= \max_j \left\{ j/n - U_{(j)}, U_{(j)} - (j-1)/n \right\}, \\ \int_{-\infty}^{\infty} \{\widehat{F}(y) - F(y)\}^2 dF(y) &= \frac{1}{12n^2} + \frac{1}{n} \sum_{j=1}^n \left(U_{(j)} - \frac{2j-1}{2n} \right)^2, \\ n \int_{-\infty}^{\infty} \frac{\{\widehat{F}(y) - F(y)\}^2}{F(y)\{1 - F(y)\}} dF(y) &= -n - \sum_{j=1}^n \frac{2j-1}{n} \log \{U_{(j)}(1 - U_{(n+1-j)})\}, \end{aligned}$$

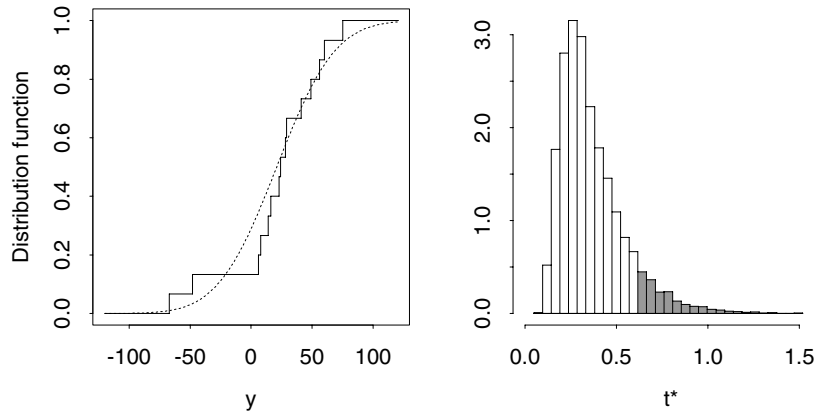
where the $U_j = F(Y_j)$ have a uniform null distribution and the $U_{(j)}$ are their order statistics; see Section 2.3. The first of these is simple and widely used, while the second and third put more weight on the tails; by allowing for the dependence of the variance of $\widehat{F}(y)$ on y , the third makes it easier to detect lack of fit for extreme values of y . All three statistics converge rapidly to their limiting distributions as $n \rightarrow \infty$, but simulation can be used to estimate P-values if tables are not at hand. The Kolmogorov–Smirnov statistic has 0.95 and 0.99 quantiles $1.358n^{-1/2}$ and $1.628n^{-1/2}$ for large n ; significance is declared if the empirical distribution function of the $U_{(j)}$ passes confidence bands defined in terms of these quantiles. See Figures 6.14 and 6.20.

Example 7.23 (Danish fire data) In Section 6.5.1 we saw that the rescaled times $u_1 = t_1/t_0, \dots, u_n = t_n/t_0$ of the events of a homogeneous Poisson process observed on $[0, t_0]$ may be regarded as the order statistics of n uniform random variables. In this case, therefore, we can take $\widehat{F}(y) = n^{-1} \sum H(y - u_j)$ and $F(y) = y$, for $0 \leq y \leq 1$, and use the above tests to assess the adequacy of the Poisson process.

$H(u)$ is the Heaviside function.

The lower right panel of Figure 6.14 shows $\widehat{F}(y)$ for the 254 largest Danish fire claims, for which the Kolmogorov–Smirnov, Cramér–von Mises, and Anderson–Darling statistics equal 0.095, 0.002, and 2.718 respectively. To assess the significance of these values we computed the three statistics for 10,000 samples of 254 independent variables generated from the $U(0, 1)$ distribution. Just 207 of the simulated Kolmogorov–Smirnov statistics exceeded the observed value, giving significance level 0.0208. The solid diagonal lines show the regions within which $\widehat{F}$ would have to fall in order for significance not to be achieved at the 0.05 and 0.01 levels, the inner 0.05 lines are breached but the outer 0.01 ones are not, consistent with significance at the 0.02 level. The significance levels for the Cramér–von Mises and Anderson–Darling statistics were 0.0348 and 0.0397, so the rate function for the claims does seem to vary. This illustrates one drawback of generic tests of fit such as these, which can suggest that the model is inadequate, but not how. ■

Figure 7.5 Analysis of maize data. Left: empirical distribution function for height differences, with fitted normal distribution (dots). Right: null density of Anderson–Darling statistic T for normal samples of size $n = 15$ with location and scale estimated. The shaded part of the histogram shows values of T^* in excess of the observed value t_{obs} .



This example is atypical, because F generally depends on unknown parameters. An exact test may be available anyway, for example using the maximal invariant of a group transformation model. An observation from a location-scale model may be written as $Y = \eta + \tau\varepsilon$, where ε has known distribution G , and $F(y) = G\{(y - \eta)/\tau\}$. Most useful estimators are equivariant, with

$$\hat{\eta}(Y_1, \dots, Y_n) = \eta + \tau h_1(\varepsilon_1, \dots, \varepsilon_n), \quad \hat{\tau}(Y_1, \dots, Y_n) = \tau h_2(\varepsilon_1, \dots, \varepsilon_n).$$

Then the joint distribution of the residuals

$$\frac{Y_j - \hat{\eta}}{\hat{\tau}} = \frac{\eta + \tau\varepsilon_j - \eta + \tau h_1(\varepsilon_1, \dots, \varepsilon_n)}{\tau h_2(\varepsilon_1, \dots, \varepsilon_n)} = \frac{\varepsilon_j - h_1(\varepsilon_1, \dots, \varepsilon_n)}{h_2(\varepsilon_1, \dots, \varepsilon_n)}, \quad j = 1, \dots, n,$$

depends only on G , h_1 , and h_2 and not on the parameters. Thus the form of G may be tested by comparing the empirical and fitted distribution functions $\hat{F}(y)$ and $G\{(y - \hat{\eta})/\hat{\tau}\}$.

Example 7.24 (Maize data) Under the matched pair model for the maize data of Table 1.1, the pairs of plants are independent and their height differences Y_j have mean η and variances $\tau = 2\sigma^2$. Our discussion in Section 3.2.2 presupposed that the Y_j are normally distributed, but the left panel of Figure 7.5 suggests that this may not be the case. To assess this we take $\hat{\eta}$ and $\hat{\tau}^2$ to be the sample average and variance, and compute the Anderson–Darling statistic based on the $(Y_j - \hat{\eta})/\hat{\tau}$. Its value is 0.618, with significance level $\hat{p}_{\text{obs}} = 0.0874$ computed from the 10,000 simulations shown in the right panel of the figure. The assumption of normality seems reasonable. ■

Similar ideas can be applied to other group transformation models. Among other goodness of fit tests are those based on the chi-squared statistics described in Section 4.5.3.

One- and two-sided tests

Often large and small values of T suggest different departures from the null hypothesis. Large values of goodness of fit statistics, for instance, imply that the model fits badly, but extremely small values might in some circumstances lead one to suspect that the

data had been faked, the fit being too good to be true. With departures of two types it may be appropriate to use T^2 or equivalently $|T|$ as the test statistic, with significance level $\Pr_0(T^2 \geq t_{\text{obs}}^2)$. This is not useful in a case like Figure 7.5, however, owing to the asymmetry of the null density of T , and then we regard the test as having two possible implications, measured by

$$p_{\text{obs}}^+ = \Pr_0(T \geq t_{\text{obs}}), \quad p_{\text{obs}}^- = \Pr_0(T \leq t_{\text{obs}}),$$

corresponding to *one-sided tests*. Note that $p_{\text{obs}}^+ + p_{\text{obs}}^- = 1 + \Pr_0(T = t_{\text{obs}})$, which equals unity if the distribution of T is continuous. Let P^+ and P^- represent the random variables corresponding to these two-sided significance levels. If both large and small values of T may be regarded as evidence against H_0 we use $P = \min(P^+, P^-)$ as the overall test statistic, and take $\Pr_0\{P \leq \min(p_{\text{obs}}^+, p_{\text{obs}}^-)\}$ as the significance level. When the test is exact and T is continuous the density of P is uniform on the interval $(0, \frac{1}{2})$, and the two-sided significance level equals $2 \min(p_{\text{obs}}^+, p_{\text{obs}}^-)$. This is the P-value for a *two-sided test*.

Example 7.25 (Student t test) Let $Y_1, \dots, Y_n$ be a normal random sample with mean μ and variance σ^2 . Suppose that the null hypothesis is $\mu = \mu_0$, and the two-sided alternative is that μ takes any other real value, with no restriction on σ^2 under either hypothesis. Both hypotheses are composite.

The likelihood ratio statistic is (Example 4.31)

$$W_p(\mu_0) = 2 \left\{ \max_{\mu, \sigma^2} \ell(\mu, \sigma^2) - \max_{\sigma^2} \ell(\mu_0, \sigma^2) \right\} = n \log \left\{ 1 + \frac{T(\mu_0)^2}{n-1} \right\},$$

where the null distribution of $T(\mu_0) = (\bar{Y} - \mu_0)/(\mathcal{S}^2/n)^{1/2}$ is t_{n-1} . As $W_p(\mu_0)$ is a monotone function of $T(\mu_0)^2$, the significance level is

$$p_{\text{obs}} = \Pr_0\{W_p(\mu_0) \geq w_{\text{obs}}\} = \Pr_0\{T^2(\mu_0) \geq t_{\text{obs}}^2\},$$

where w_{obs} and t_{obs} are the observed values of $W_p(\mu_0)$ and $T(\mu_0)$. Large values of w_{obs} arise when t_{obs} is distant from zero, suggesting that the population mean is not μ_0 .

The results of Section 4.5 tell us that the null distribution of $W_p(\mu_0)$ is approximately χ_1^2 . We could use this to approximate to p_{obs} , but an exact value is available, because

$$p_{\text{obs}} = \Pr_0\{T(\mu_0)^2 \geq t_{\text{obs}}^2\} = \Pr(T^2 \geq t_{\text{obs}}^2) = 2\Pr(T \geq |t_{\text{obs}}|), \quad (7.27)$$

where $T \sim t_{n-1}$. This is the P-value for the two-sided test.

If we suspect that $\mu > \mu_0$ but not that $\mu < \mu_0$, then large positive values of $T(\mu_0)$ will cast doubt on H_0 , and the corresponding one-sided P-value is

$$p_{\text{obs}}^+ = \Pr_0\{T(\mu_0) \geq t_{\text{obs}}\} = \Pr(T \geq t_{\text{obs}}),$$

while $p_{\text{obs}}^- = \Pr(T \leq t_{\text{obs}})$ measures evidence against H_0 in the direction $\mu < \mu_0$. These differ slightly from the P-values for the one-sided likelihood ratio tests. The

two-sided significance level

$$2 \min(p_{\text{obs}}^-, p_{\text{obs}}^+) = 2\Pr(|T| \geq |t_{\text{obs}}|)$$

equals (7.27). ■

Nonparametric tests

The examples above concern tests in parametric models, where hypotheses typically determine values of the parameters, the form of the density being supposed known. *Nonparametric tests* presuppose that the data are independently sampled from an unspecified underlying model.

Example 7.26 (Sign test) A random sample $Y_1, \dots, Y_n$ arises from an unknown distribution F . The null hypothesis H_0 asserts that F has median μ equal to μ_0 , while the alternative is that $\mu > \mu_0$. Both hypotheses are composite, but neither specifies a parametric model, and we argue as follows.

If the median is μ_0 , the probability that an observation Y falls on either side of μ_0 is $1/2$, and if the median is greater than μ_0 , then $\Pr(Y > \mu_0) > 1/2$. This suggests that we base a test on $S = \sum_{j=1}^n I(Y_j > \mu_0)$, large values of which cast doubt on H_0 . Under the null hypothesis, S has a binomial distribution with denominator n and probability $1/2$, so its mean and variance are $n/2$ and $n/4$. Hence the P-value is

$$p_{\text{obs}} = \Pr_0(S \geq s_{\text{obs}}) = \sum_{r=s_{\text{obs}}}^n \binom{n}{r} \frac{1}{2^n} \doteq 1 - \Phi \left\{ \frac{2(s_{\text{obs}} - n/2)}{n^{1/2}} \right\},$$

by normal approximation to the binomial null distribution of S . ■

Example 7.27 (Wilcoxon signed-rank test) A random sample $Y_1, \dots, Y_n$ has been drawn from a density that is symmetric about μ but otherwise unspecified. We wish to test the hypothesis that $\mu = 0$. The sign test is one possibility, but as it does not use the symmetry of the density, a better test can be found.

Let R_j denote the rank of $|Y_j|$ among $|Y_1|, \dots, |Y_n|$, and let $Z_j = \text{sign}(Y_j)$. The *Wilcoxon signed-rank statistic* is $W = \sum_j Z_j R_j$. Large positive values of W suggest $\mu > 0$, while large negative values suggest $\mu < 0$.

To find the null mean and variance of W , note that when $\mu = 0$ the ranks, R_j , are independent of the signs, Z_j , by symmetry about zero, and that

$$\text{var}_0(Z_j) = (-1)^2 \frac{1}{2} + 1^2 \frac{1}{2} = 1, \quad \text{E}_0(Z_j R_j) = n^{-1} \sum_{k=1}^n \left\{ k \frac{1}{2} + (-k) \frac{1}{2} \right\} = 0,$$

implying that $\text{E}_0(W) = 0$. To find $\text{var}_0(W)$, we argue conditionally on the ranks $R_1, \dots, R_n$, finding

$$\text{var}_0 \left(\sum_{j=1}^n Z_j R_j \middle| R_1, \dots, R_n \right) = \sum_{j=1}^n R_j^2 \text{var}_0(Z_j) = \sum_{j=1}^n R_j^2 = \sum_{j=1}^n j^2,$$

and this equals $n(n+1)(2n+1)/6$. Thus W has mean zero and variance $n(n+1)(2n+1)/6$ under the null hypothesis, and as its distribution is then symmetric, a normal approximation to the exact P-value may be useful. ■

Difference d	49	-67	8	16	6	23	28	41	14	29	56	24	75	60	-48
Sign z	+	-	+	+	+	+	+	+	+	+	+	+	+	+	-
Rank r	11	14	2	4	1	5	7	9	3	8	12	6	15	13	10

Table 7.2 Analysis of differences for maize data.

Example 7.28 (Maize data) Under the model for the maize data of Table 1.1, the height differences between cross- and self-fertilized plants may be written as $D_j = \eta + \sigma(\varepsilon_{2j} - \varepsilon_{1j})$, where the ε_{ij} are independent random variables with mean zero and some common variance. If the ε_{ij} have the same distribution, the D_j will be symmetrically distributed around η , while $\eta = 0$ under the null hypothesis H_0 of no difference between the effects of the different types of fertilization. If cross-fertilization increases height, then $\eta > 0$, as is suggested by the observed d_j in Table 7.2.

If the D_j were normally distributed, we would perform a Student t test based on the average and variance of the observed differences, $\bar{d} = 20.95$ and $s^2 = 1424.6$, giving $t_{\text{obs}} = n^{1/2}(\bar{d} - 0)/s = 2.15$; see Example 7.25. Under H_0 this is the realized value of a t_{14} variable, so $p_{\text{obs}} = \Pr(T \geq t_{\text{obs}}) = 0.025$, where $T \sim t_{14}$. Though low, this is not overwhelming evidence against the null hypothesis.

If we wish to avoid the assumption of normality, a nonparametric test is preferable. Under the null hypothesis, the D_j come from density symmetric about zero but not necessarily normal. Thirteen of them are positive, so the sign test statistic takes value $s_{\text{obs}} = 13$, with exact significance level

$$\Pr_0(S \geq s_{\text{obs}}) = \frac{1}{2^{15}} \sum_{r=13}^{15} \binom{15}{r} = \frac{1}{2^{15}} (1 + 15 + 105) = 0.0037;$$

normal approximation gives $1 - \Phi\{2(13 - 15/2)/\sqrt{15}\} = 0.0023$. Both give much stronger evidence against H_0 than does the t test.

Table 7.2 shows the quantities needed for the Wilcoxon signed-rank test. The observed value of $W = \sum Z_j R_j$ is 72, and its null distribution when $n = 15$ is approximately normal with mean zero and variance 1240. Therefore the P-value is roughly

$$p_{\text{obs}} = \Pr_0(W \geq 57) \doteq 1 - \Phi(57/1240^{1/2}) = 0.053,$$

to be compared with the values for the t and sign tests. ■

We shall see in Section 7.3.2 that likelihood considerations lead to tests that are ‘best’ in a certain sense when there is a parametric model. But if the model is not credible, nonparametric tests that make fewer assumptions may be preferable, and often they perform nearly as well as parametric tests. Some situations are so ill-specified that parametric models are inappropriate, and the independence assumptions that underlie most nonparametric tests are doubtful also. Then only rough-and-ready methods can be applied and conclusions are correspondingly weaker.

7.3.2 Comparison of tests

We now consider how to compare different test statistics for the same problem. Having chosen a test statistic $T = t(Y)$ and a probability α , suppose we decide to reject the null hypothesis H_0 in favour of an alternative H_1 at level α if and only if the data Y fall into the subset $\mathcal{Y}_\alpha = \{y : t(y) \geq t_\alpha\}$ of the sample space, where t_α is chosen so that

$$\Pr_0(T \geq t_\alpha) = \Pr_0(Y \in \mathcal{Y}_\alpha) = \alpha.$$

The *size* of the test is the probability α of rejecting H_0 when it is actually true, and $\mathcal{Y}_\alpha$ is called a *size α critical region*. This construction implies that as α decreases, t_α increases and that $\mathcal{Y}_{\alpha_1} \subset \mathcal{Y}_{\alpha_2}$ whenever $\alpha_1 \leq \alpha_2$, as is essential if we are to avoid imbecilities such as ‘ H_0 is rejected when $\alpha = 0.01$ but not when $\alpha = 0.05$ ’. Choosing a test statistic and values of t_α is equivalent to specifying a system of critical regions for the different values of α , so we can discuss the test in terms of its critical regions if convenient.

By using a fixed α we have moved from regarding the significance level as a measure of evidence against H_0 to using the test to decide which of the two hypotheses is better supported by the data. Two wrong decisions are then possible, committing a *Type I* error by rejecting H_0 when it is true, or a *Type II* error by accepting H_0 when H_1 is true. The *power* of the test is the probability of detecting that H_0 is false,

$$\Pr_1(T \geq t_\alpha) = \Pr_1(Y \in \mathcal{Y}_\alpha).$$

Example 7.29 (Normal mean) Let $Y_1, \dots, Y_n$ be a random sample from the $N(\mu, \sigma^2)$ distribution with known σ^2 , and suppose that H_0 specifies that $\mu = \mu_0$, whereas $\mu > \mu_0$ under H_1 . Suppose we decide to reject H_0 if $\bar{Y}$ exceeds some constant t_α . Under H_0 , $\bar{Y} \sim N(\mu_0, \sigma^2/n)$, so this test has size

$$\begin{aligned} \Pr_0(\bar{Y} \geq t_\alpha) &= \Pr_0 \left\{ n^{1/2} \frac{(\bar{Y} - \mu_0)}{\sigma} \geq n^{1/2} \frac{(t_\alpha - \mu_0)}{\sigma} \right\} \\ &= 1 - \Phi \left\{ \frac{n^{1/2}(t_\alpha - \mu_0)}{\sigma} \right\} = \Phi \left\{ \frac{n^{1/2}(\mu_0 - t_\alpha)}{\sigma} \right\}, \end{aligned}$$

using the symmetry of the normal distribution. For a test of size α , we must choose t_α such that

$$\frac{n^{1/2}(\mu_0 - t_\alpha)}{\sigma} = z_\alpha,$$

giving $t_\alpha = \mu_0 - n^{-1/2}\sigma z_\alpha$. Thus the size α critical region is

$$\mathcal{Y}_\alpha = \{(y_1, \dots, y_n) : \bar{y} \geq \mu_0 - n^{-1/2}\sigma z_\alpha\},$$

and we can decide if Y falls into this because σ^2 and μ_0 are known under H_0 .

$\Pr_1, E_1$ and so forth indicate probability, expectation and so forth computed under H_1 .

z_α is the α quantile of the $N(0, 1)$ distribution.

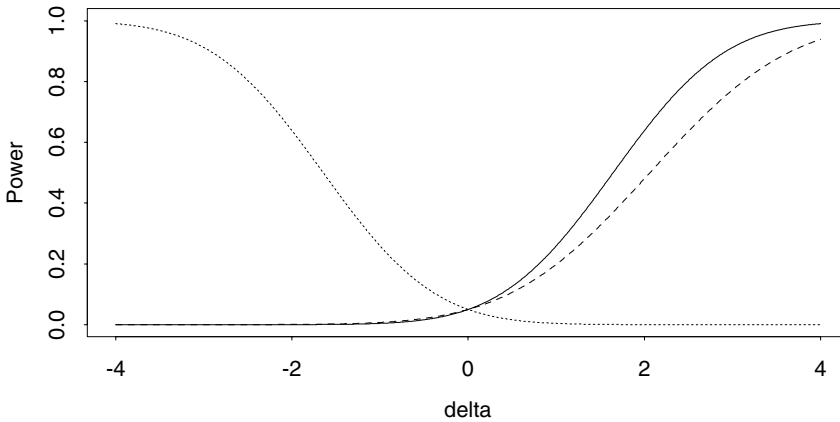


Figure 7.6 Power functions for a test of whether the mean of a $N(\mu, \sigma^2)$ random sample of size n equals μ_0 against the alternative $\mu = \mu_1$, as a function of $\delta = n^{1/2}(\mu_1 - \mu_0)/\sigma$. The test size is $\alpha = 0.05$. The solid curve is the power function for a test of $\mu_1 > \mu_0$ based on $\bar{y}$, and the dashed line is the power function for the sign test. Both critical regions are of form $\bar{y} > t_\alpha$. The dotted curve is the power function for $\bar{y}$ when the critical region is $\bar{y} < t_\alpha$.

If in fact μ equals $\mu_1 > \mu_0$, then $\bar{Y} \sim N(\mu_1, \sigma^2/n)$, and the test has power

$$\begin{aligned} \Pr_1\left(\bar{Y} \geq \mu_0 - \frac{\sigma z_\alpha}{n^{1/2}}\right) &= \Pr_1\left(n^{1/2} \frac{(\bar{Y} - \mu_1)}{\sigma} \geq n^{1/2} \frac{(\mu_0 - \mu_1)}{\sigma} - z_\alpha\right) \\ &= 1 - \Phi(-\delta - z_\alpha) = \Phi(z_\alpha + \delta), \end{aligned} \quad (7.28)$$

where $\delta = n^{1/2}(\mu_1 - \mu_0)/\sigma$ measures the distance between the means under the two hypotheses, standardized by $\text{var}(\bar{Y})^{1/2} = \sigma/n^{1/2}$. The power is plotted in Figure 7.6, with $\alpha = 0.05$. For fixed n , σ , and μ_0 , it increases with μ_1 . When σ , μ_0 , and μ_1 are fixed, the power increases with n .

Power can be used to choose the sample size when planning an experiment. Suppose we desire to perform a test of size α and that power of at least β is sought for detecting whether $\mu_1 = \mu_0 + \sigma\gamma$, where γ is known. Then we require $\Phi(z_\alpha + n^{1/2}\gamma) \geq \beta$ and hence $z_\alpha + n^{1/2}\gamma \geq \Phi^{-1}(\beta)$ or equivalently $n \geq (z_\beta - z_\alpha)^2/\gamma^2$.

If, for instance, $\mu_0 = 0$ and $\sigma = 1$, and we desire to detect whether a test of size 0.05 could detect $\mu_1 = 0.5$ with power 0.8 or more, then $\gamma = 0.5$, $z_\alpha = -1.645$, $z_\beta = 0.842$ and hence we would need $n \geq 24.7 \doteq 25$. ■

Example 7.30 (Sign test) Example 7.26 describes a test for the median of a distribution to equal a specified value μ_0 , using $S = \sum_{j=1}^n I(Y_j > \mu_0)$ as test statistic. Under H_0 the distribution of S is binomial, and if a normal approximation applies, a size α critical region is determined by the value s_α such that $\Pr_0(S \geq s_\alpha) = \alpha$, giving $s_\alpha = n/2 - n^{1/2}z_\alpha/2$.

For an illustrative power calculation for this test, let $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, with null hypothesis $\mu = \mu_0$ and alternative H_1 that $\mu = \mu_1 > \mu_0$. The normal density is symmetric, so its mean equals its median. Now

$$\Pr_1(Y_j \geq \mu_0) = \Pr_1\{(Y_j - \mu_1)/\sigma \geq (\mu_0 - \mu_1)/\sigma\} = \Phi(n^{-1/2}\delta),$$

where again $\delta = n^{1/2}(\mu_1 - \mu_0)/\sigma$. Under H_1 , therefore, S is approximately normal with mean $n\Phi(n^{-1/2}\delta)$ and variance $n\Phi(n^{-1/2}\delta)\{1 - \Phi(n^{-1/2}\delta)\}$, and the probability

that H_0 is rejected is

$$\begin{aligned}\Pr_1(S \geq s_\alpha) &= \Pr_1(S \geq n/2 - n^{1/2}z_\alpha/2) \\ &\doteq \Phi \left\{ \frac{n\Phi(n^{-1/2}\delta) - n/2 + n^{1/2}z_\alpha/2}{[n\Phi(n^{-1/2}\delta) \{1 - \Phi(n^{-1/2}\delta)\}]^{1/2}} \right\},\end{aligned}$$

using the normal approximation to the binomial distribution. For n large, $\Phi(n^{-1/2}\delta) \doteq \frac{1}{2} + n^{-1/2}\delta\phi(0) = \frac{1}{2} + (2\pi n)^{-1/2}\delta$, and after simplifying,

$$\Pr_1(S \geq s_\alpha) \doteq \Phi\{z_\alpha + \delta(2/\pi)^{1/2}\}. \quad (7.29)$$

As $(2/\pi)^{1/2} < 1$, the sign test has lower power than does the test using $\bar{Y}$ in Example 7.29. That test has power $\Phi(z_\alpha + \delta)$, so it requires smaller samples to attain a given power than does the test based on S . Figure 7.6 compares the power functions with $\alpha = 0.05$. Sign tests have rather low power, and better tests are almost always possible. ■

Although power is important in planning an experiment, in giving a basis for choosing the sample size required, and in assessing the size of effects that could reasonably be detected from a given set of data, it plays no role in conducting the test itself, which simply requires a tail probability computed under the null distribution.

Neyman–Pearson lemma

Other things being equal, a test with high power is preferable to one with low power. But in order for a comparison of two tests to be fair, they must compete on an equal footing. This leads us to compare them in terms of their power for fixed size. That is, out of all possible tests with a given size, we aim to find the one with highest power.

Let $f_0(y)$ and $f_1(y)$ denote the probability densities of Y under the null and alternative hypotheses. Then the *Neyman–Pearson lemma* states that the most powerful test of size α has critical region

$$\mathcal{Y} = \left\{ y : \frac{f_1(y)}{f_0(y)} \geq t_\alpha \right\}, \quad t_\alpha \geq 0,$$

determined by the likelihood ratio, if such a region exists. To explain this, suppose that such a region does exist and let $\mathcal{Y}'$ be any other critical region of size α or less. Then for any density f ,

$$\int_{\mathcal{Y}} f(y) dy - \int_{\mathcal{Y}'} f(y) dy,$$

equals

$$\int_{\mathcal{Y} \cap \mathcal{Y}'} f(y) dy + \int_{\mathcal{Y} \cap \bar{\mathcal{Y}}'} f(y) dy - \int_{\mathcal{Y}' \cap \mathcal{Y}} f(y) dy - \int_{\mathcal{Y}' \cap \bar{\mathcal{Y}}} f(y) dy,$$

and this is

$$\int_{\mathcal{Y} \cap \bar{\mathcal{Y}}'} f(y) dy - \int_{\mathcal{Y}' \cap \bar{\mathcal{Y}}} f(y) dy. \quad (7.30)$$

Egon Sharpe Pearson (1895–1980), the second child of Karl Pearson, was very unlike his combative father. After school in Oxford and Winchester his studies in Cambridge were interrupted by illness and the 1914–18 war. He took his degree in 1920 and began work at University College London, where he stayed the rest of his life. Apart from broad contributions to statistical theory, he pioneered industrial quality control and was editor of the statistical journal *Biometrika* from 1936–1966.

$\bar{\mathcal{Y}}$ is the complement of $\mathcal{Y}$ in the sample space.

If $f = f_0$, this expression is non-negative, because $\mathcal{Y}'$ has size at most that of $\mathcal{Y}$. Suppose that $f = f_1$. If $y \in \overline{\mathcal{Y}}$, then $t_\alpha f_0(y) > f_1(y)$, while $f_1(y) \geq t_\alpha f_0(y)$ if $y \in \mathcal{Y}$. Hence when $f = f_1$, (7.30) is no smaller than

$$t_\alpha \left\{ \int_{\mathcal{Y} \cap \overline{\mathcal{Y}'}} f_0(y) dy - \int_{\mathcal{Y}' \cap \overline{\mathcal{Y}}} f_0(y) dy \right\} \geq 0.$$

Thus the power of $\mathcal{Y}$ is at least that of $\mathcal{Y}'$, and the result is established.

It may happen that H_0 is simple and the alternative is composite, but that the likelihood ratio critical region is most powerful for each component of the alternative hypothesis. Then $\mathcal{Y}$ is said to be *uniformly most powerful*.

Example 7.31 (Exponential family) Consider testing the null hypothesis $\theta = \theta_0$ against the one-sided alternative $\theta = \theta_1 > \theta_0$ based on a random sample $Y_1, \dots, Y_n$ from the one-parameter exponential family

$$f(y; \theta) = \exp \{s(y)\theta - \kappa(\theta) + c(y)\}.$$

The likelihood ratio is

$$\exp \left\{ (\theta_1 - \theta_0) \sum_{j=1}^n s(Y_j) + \kappa(\theta_0) - \kappa(\theta_1) \right\},$$

so for each $\theta_1 > \theta_0$ the most powerful size α critical region is

$$\mathcal{Y}_\alpha = \left\{ (y_1, \dots, y_n) : \sum s(y_j) \geq t'_\alpha \right\},$$

if a t'_α can be found such that $\Pr_0(Y \in \mathcal{Y}_\alpha) = \alpha$. This test is therefore uniformly most powerful against this one-sided alternative. When $\theta_1 < \theta_0$, the same argument shows that a uniformly most powerful critical region is obtained by replacing $\geq$ by $\leq$ in the above definition of $\mathcal{Y}_\alpha$.

A special case of this is the exponential density of Example 7.22, where the uniformly most powerful critical region of size α against one-sided alternatives $\lambda_1 < \lambda_0$ is $\mathcal{Y}_\alpha = \{(y_1, \dots, y_n) : \sum y_j > t'_\alpha\}$, with $\lambda_0 t'_\alpha$ the $(1 - \alpha)$ quantile of the gamma distribution with unit scale and shape parameter n .

In discrete models uniformly most powerful tests of every size do not exist. In the Poisson case, for example, the null distribution of $\sum s(Y_j) = \sum Y_j$ is Poisson with mean $n\theta_0$, so $\mathcal{Y}_\alpha$ has possible sizes

$$\Pr_0 \left(\sum_{j=1}^n Y_j \geq t'_\alpha \right) = \sum_{u=t'_\alpha}^{\infty} \frac{(n\theta_0)^u}{u!} \exp(-n\theta_0), \quad t_\alpha = 0, 1, \dots$$

Setting $n\theta_0 = 5$, for example, gives sizes 1.00, 0.993, ..., 0.068, 0.032, ..., so a likelihood ratio critical region of size 0.05 does not exist. This does not affect the computation of a significance level, whose value is not pre-specified. ■

This last example shows that construction of a likelihood ratio critical region of exact size α may be impossible. If so, a *randomized test* may be used to obtain the exact size required. Suppose that critical regions of size α_1 and α_2 are available,

where $\alpha_1 < \alpha < \alpha_2$. Then if I is a Bernoulli variable with success probability $p = (\alpha_2 - \alpha)/(\alpha_2 - \alpha_1)$, the test with region

$$\mathcal{Y} = \begin{cases} \mathcal{Y}_{\alpha_1}, & I = 1, \\ \mathcal{Y}_{\alpha_2}, & I = 0 \end{cases}$$

has size α . In the previous example we might take $\alpha = 0.05$, $\alpha_1 = 0.032$ and $\alpha_2 = 0.068$, giving $p = 0.5$. Then each time the test was conducted, we would flip a coin to decide whether to use $\mathcal{Y}_{\alpha_1}$ or $\mathcal{Y}_{\alpha_2}$ as the critical region. Although this trick is useful in theoretical calculations, it introduces a random element unrelated to the data. In applications it is preferable to compute a significance level and weigh the evidence accordingly.

Example 7.32 (Normal mean) In Example 7.29 the likelihood ratio for testing $\mu = \mu_0$ against $\mu = \mu_1$ with σ known is

$$\begin{aligned} \frac{f_1(Y)}{f_0(Y)} &= \frac{(2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{j=1}^n (Y_j - \mu_1)^2\right\}}{(2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{j=1}^n (Y_j - \mu_0)^2\right\}} \\ &= \exp\left[\frac{1}{2\sigma^2} \{2n\bar{Y}(\mu_1 - \mu_0) - \mu_1^2 + \mu_0^2\}\right]. \end{aligned}$$

If $\mu_1 > \mu_0$, this is monotone increasing in $\bar{Y}$ for any fixed μ_1 and μ_0 , and so the critical region rejects H_0 when $\bar{Y} \geq t'_\alpha$, with t'_α chosen to give a test of size α . Hence the size α critical region is

$$\mathcal{Y}_\alpha^+ = \{(y_1, \dots, y_n) : n^{1/2}(\bar{y} - \mu_0)/\sigma \geq z_{1-\alpha}\};$$

this is most powerful for any $\mu_1 > \mu_0$ and so is uniformly most powerful. The region

$$\mathcal{Y}_\alpha^- = \{(y_1, \dots, y_n) : n^{1/2}(\bar{y} - \mu_0)/\sigma \leq z_\alpha\}$$

is likewise uniformly most powerful against alternatives $\mu_1 < \mu_0$.

Suppose that we wish to test the same null hypothesis against the two-sided alternative that $\mu \neq \mu_0$. The null distribution of $\bar{Y}$ is symmetric about μ_0 , so it is natural to use

$$\mathcal{Y}_\alpha = \{(y_1, \dots, y_n) : n^{1/2}|\bar{y} - \mu_0|/\sigma \geq z_{\alpha/2}\}. \quad (7.31)$$

This critical region has size α but is not uniformly most powerful against the two-sided alternative. When $\mu_1 > \mu_0$, $\mathcal{Y}_\alpha^+$ has size α and has higher power, while when $\mu_1 < \mu_0$, $\mathcal{Y}_\alpha^-$ has size α and has higher power. The power of a uniformly most powerful two-sided critical region would equal those of $\mathcal{Y}_\alpha^+$ for alternatives $\mu_1 > \mu_0$ and of $\mathcal{Y}_\alpha^-$ for $\mu_1 < \mu_0$, but its size would have to be α , whereas $\mathcal{Y}_\alpha^- \cup \mathcal{Y}_\alpha^+$ has size 2α . In fact no uniformly most powerful test exists for this two-sided alternative. This difficulty can also arise in other contexts. ■

This last example highlights a problem with two-sided tests. One approach to dealing with it is to say that a critical region $\mathcal{Y}$ is *unbiased* if

$$\Pr_1(Y \in \mathcal{Y}) \geq \Pr_0(Y \in \mathcal{Y})$$

for all alternative hypotheses under consideration. This implies that the probability of rejecting H_0 is higher under any H_1 than under H_0 , and would rule out using the critical regions $\mathcal{Y}_\alpha^+$ and $\mathcal{Y}_\alpha^-$ for two-sided tests in the previous example. If $\mu_1 < \mu_0$, for example, then $\Pr_1(Y \in \mathcal{Y}_\alpha^+) = \Phi(z_\alpha + \delta) < \alpha$ because $\delta < 0$, and hence $\mathcal{Y}_\alpha^+$ would be biased. There is a well-developed mathematical theory of such tests, but they are of little practical interest. To see why, suppose that the two-sided unbiased region $\mathcal{Y}_\alpha$ had been used in the previous example, and that doubt had been cast on the null hypothesis $\mu = \mu_0$. The test being two-sided, it would then be natural to ask whether the data suggest that $\mu > \mu_0$ or $\mu < \mu_0$, leading to use of one-sided regions such as $\mathcal{Y}_\alpha^-$ and $\mathcal{Y}_\alpha^+$. It seems more sensible to perform two one-sided tests and obtain an overall P-value by combining the individual significance levels, as outlined in Section 7.3.1. This amounts to using two one-sided tests each of size α , and in general this is not the same as an unbiased test of size 2α .

Local power

We now consider how the likelihood ratio behaves under a local alternative, when the null and alternative models $f_0(y) = f(y; \theta_0)$ and $f_1(y) = f(y; \theta_1)$ depend on a scalar parameter θ , and $\theta_1 = \theta_0 + \epsilon$ for some small ϵ . Then

$$\begin{aligned} \frac{f_1(Y)}{f_0(Y)} &= \frac{f(Y; \theta_0 + \epsilon)}{f(Y; \theta_0)} = \frac{1}{f(Y; \theta_0)} \left\{ f(Y; \theta_0) + \epsilon \frac{df(Y; \theta_0)}{d\theta_0} + \dots \right\} \\ &\doteq 1 + \epsilon U(\theta_0), \end{aligned}$$

where $U(\theta) = d \log f(Y; \theta) / d\theta$ is the score statistic. As $\epsilon \rightarrow 0$, this expansion shows that the likelihood ratio and score statistics are equivalent, so the Neyman–Pearson lemma implies that a locally most powerful test against H_0 may be based on large values of the score statistic. This is a *score test*.

In large samples from regular models the null distribution of $U(\theta_0)$ is approximately normal with mean zero and variance equal to the Fisher information $I(\theta_0)$, so a locally most powerful critical region has form

$$\{(y_1, \dots, y_n) : u(\theta_0) \geq I(\theta_0)^{1/2} z_{1-\alpha}\}.$$

Under the alternative hypothesis, $U(\theta_0)$ has mean

$$\begin{aligned} \int u(\theta_0) f(y; \theta_0 + \epsilon) dy &= \int u(\theta_0) \{f(y; \theta_0) + \epsilon u(\theta_0) f(y; \theta_0) + \dots\} dy \\ &\doteq \epsilon \int u(\theta_0)^2 f(y; \theta_0) dy = \epsilon I(\theta_0), \end{aligned}$$

while its variance is $I(\theta_0) + O(n\epsilon)$. Hence the local power of the score test is

$$\Pr_1 \{U(\theta_0) \geq I(\theta_0)^{1/2} z_{1-\alpha}\} \doteq \Phi(z_\alpha + \delta),$$

analogous to (7.28), with $\delta = I(\theta_0)^{1/2}(\theta_1 - \theta_0) = n^{1/2}(\theta_1 - \theta_0)/i(\theta_0)^{-1/2}$ playing the role of $n^{1/2}(\mu_1 - \mu_0)/\sigma$ in Example 7.29. Thus the power of the test is increased when the null Fisher information per observation $i(\theta_0)$ is large, when n is large, or when θ_1 is distant from θ_0 .

Example 7.33 (Gamma density) Suppose that $Y_1, \dots, Y_n$ is a random sample from the gamma density

$$f(y; \mu, \nu) = \frac{\nu^\nu y^{\nu-1}}{\Gamma(\nu)\mu^\nu} \exp(-\nu y/\mu), \quad y > 0, \nu, \mu > 0.$$

We consider testing if $\nu = 1$, that is, that the density is in fact exponential. Initially we suppose that μ is known. The log likelihood contribution from a single observation is $\nu \log \nu + (\nu - 1) \log y - \nu y/\mu - \log \Gamma(\nu)$, so

$$U(\nu) = \sum_{j=1}^n \left\{ \log \left(\frac{Y_j}{\mu} \right) - \frac{Y_j}{\mu} + 1 - \log \nu - \frac{d \log \Gamma(\nu)}{d\nu} \right\},$$

$$I(\nu) = n \left\{ \frac{d^2 \log \Gamma(\nu)}{d\nu^2} - \frac{1}{\nu} \right\}.$$

An asymptotic test of $\nu = 1$ therefore consists in comparing $U(1)/I(1)^{1/2}$ with the standard normal distribution.

In practice an unknown μ is replaced by its maximum likelihood estimator under the null hypothesis, $\hat{\mu} = \bar{Y}$. Then the large-sample distribution of the score is given by (4.48) with $\psi = \nu$ and $\lambda = \mu$. In this case the off-diagonal element of the Fisher information matrix is $I_{\lambda,\psi} = E(-\partial^2 \ell / \partial \mu \partial \nu) = 0$, so the test involves replacing μ by $\bar{Y}$. ■

7.3.3 Composite null hypotheses

Thus far we have supposed that the null hypothesis is simple, that is, it fully specifies the null distribution of the test statistic. An exact significance level, perhaps estimated by simulation, is then in principle available. In practice exact tests are usually unobtainable because the null distribution of Y depends on unknowns. In the most common setting there is a nuisance parameter λ and a parameter of interest ψ , and the null hypothesis imposes the constraint $\psi = \psi_0$ but puts no restriction on λ . Most of the tests in preceding chapters were of this sort. The P-value may then be written

$$\Pr_0(T \geq t_{\text{obs}}) = \Pr(T \geq t_{\text{obs}}; \psi_0, \lambda) = \int_{\{y: t(y) \geq t_{\text{obs}}\}} f(y; \psi_0, \lambda) dy. \quad (7.32)$$

In general this depends on λ , perhaps strongly, but sometimes a critical region $\mathcal{Y}_\alpha$ of size α can be found such that

$$\Pr(Y \in \mathcal{Y}_\alpha; \psi_0, \lambda) = \alpha \quad \text{for all } \lambda.$$

Such a $\mathcal{Y}_\alpha$ is called a *similar region*; it is similar to the sample space, which satisfies this equation with $\alpha = 1$. A test whose critical regions are similar is called a *similar test* and is clearly desirable if it can be found. The two main approaches to finding exact tests are use of conditioning and appeal to invariance. Before discussing these, we outline approximate ways to reduce the dependence of (7.32) on λ .

One simple idea is to replace λ by $\hat{\lambda}_0$, the maximum likelihood estimator of λ when $\psi = \psi_0$, but this is generally unsatisfactory because the result still depends on λ , albeit

to a lower order. It is better to base the test on a pivot, exact or approximate. We have already extensively used an important example of this, the likelihood ratio statistic $W_p(\psi_0) = 2\{\ell(\hat{\psi}, \hat{\lambda}) - \ell(\psi_0, \hat{\lambda}_0)\}$. Under regularity conditions its distribution for a large sample size n is χ_p^2 , where p is the dimension of ψ , and in fact as

$$\Pr\{W_p(\psi_0) \leq c_p(\alpha); \psi_0, \lambda\} = \alpha\{1 + O(n^{-1})\} \quad \text{for all } \lambda, \quad (7.33)$$

tests based on $W_p(\psi_0)$ are approximately similar. In continuous models the error in (7.33) can be reduced by noting that $E_0\{W_p(\psi_0)\} \doteq p\{1 + b(\theta_0)/n\}$, where $b(\theta_0) = b(\psi_0, \lambda)$ conveys how much the null mean of $W_p(\psi_0)$ differs from its asymptotic value. Tedious calculations establish that

$$\Pr\{W_p(\psi_0)\{1 + b(\hat{\theta}_0)\}^{-1} \leq c_p(\alpha); \psi_0, \lambda\} = \alpha\{1 + O(n^{-2})\} \quad \text{for all } \lambda,$$

where $\hat{\theta}_0 = (\psi_0, \hat{\lambda}_0)$. Thus division of the likelihood ratio statistic to make its mean closer to p improves the quality of the χ^2 approximation to its entire distribution. *Bartlett adjustment* of this sort can decrease substantially the error in (7.33), and may be valuable if n is small or if the dimension of λ is appreciable.

Conditioning

When there is a minimal sufficient statistic S_0 for the unknown λ in a null distribution, it may be removed by conditioning, giving P-value

$$\Pr_0(T \geq t_{\text{obs}} \mid S_0; \psi_0) = \int_{\{y: t(y) \geq t_{\text{obs}}\}} f(y \mid s_0; \psi_0) dy,$$

which is independent of λ by sufficiency of S_0 . If S_0 is boundedly complete, this is the only way to construct a test statistic with P-values independent of λ . To see why, let $\mathcal{Y}_\alpha$ be a critical region of size α for all λ . Then

$$\begin{aligned} 0 &= \Pr_0(Y \in \mathcal{Y}_\alpha; \psi_0, \lambda) - \alpha = E\{I(Y \in \mathcal{Y}_\alpha) - \alpha; \psi_0, \lambda\} \\ &= E_{S_0}[E\{I(Y \in \mathcal{Y}_\alpha) \mid S_0; \psi_0\} - \alpha; \psi_0, \lambda], \end{aligned}$$

for all λ , and the bounded completeness of S_0 implies that

$$E\{I(Y \in \mathcal{Y}_\alpha) \mid S_0; \psi_0\} = \Pr(Y \in \mathcal{Y}_\alpha \mid S_0; \psi_0) = \alpha.$$

Hence similar critical regions must be based on this conditional density.

Example 7.34 (Exponential family) In Section 5.2.3 we saw that conditioning on the statistic S_2 associated with λ in the full exponential family model

$$f(s_1, s_2; \psi, \lambda) = \exp\{s_1^T \psi + s_2^T \lambda - \kappa(\psi, \lambda)\} g_0(s_1, s_2),$$

gives a density independent of λ , namely

$$f(s_1 \mid s_2; \psi) = \exp\{s_1^T \psi - \kappa_{s_2}(\psi)\} g_{s_2}(s_1). \quad (7.34)$$

If a particular value ψ_0 of ψ is fixed, then S_2 is complete and minimal sufficient for λ . Hence similar critical regions for testing $\psi = \psi_0$ must be based on (7.34).

Consider two independent Poisson variables with means μ_1 and μ_2 , and suppose that we wish to test the hypothesis $\mu_1 = \mu_2$. We may equivalently set

$c_p(\alpha)$ is the α quantile of the χ_p^2 distribution.

Maurice Stevenson Bartlett (1910–2002) worked at research institutes and the universities of London, Manchester, and Oxford. Starting in the mid 1930s, he made pioneering contributions to likelihood inference, to multivariate analysis and to stochastic processes, on which he wrote a highly influential book.

$\mu_1 = \exp(\lambda + \psi)$ and $\mu_2 = \exp(\lambda)$ with $-\infty < \psi, \lambda < \infty$ and test the hypothesis $\psi = 0$ with no restriction on λ . The corresponding exponential family model is

$$\frac{\mu_1^{y_1}}{y_1!} e^{-\mu_1} \times \frac{\mu_2^{y_2}}{y_2!} e^{-\mu_2} = \frac{1}{y_1! y_2!} \exp\{y_1 \psi + (y_1 + y_2) \lambda - e^{\lambda + \psi} - e^\lambda\},$$

where $y_1, y_2 \in \{0, 1, \dots\}$. Here $S_2 = Y_1 + Y_2$ has a Poisson distribution with mean $\mu_1 + \mu_2 = e^\lambda(1 + e^\psi)$, so the conditional density of $S_1 = Y_1$ is binomial,

$$f(s_1 | s_2; \psi) = \frac{s_2!}{s_1!(s_2 - s_1)!} \left(\frac{e^\psi}{1 + e^\psi} \right)^{s_1} \left(\frac{1}{1 + e^\psi} \right)^{s_2 - s_1}, \quad s_1 = 0, 1, \dots, s_2.$$

This has denominator $s_2 = y_1 + y_2$ and so treats the total for the two variables as fixed. When $\psi = 0$ the probability equals $1/2$, so the only similar critical regions for a test of $\psi = 0$ against $\psi > 0$, that is, $\mu_1 > \mu_2$, have form

$$\Pr_0(Y_1 \geq r' | Y_1 + Y_2 = s_2) = \sum_{r=r'}^{s_2} \binom{s_2}{r} 2^{-r}, \quad r' = 0, 1, \dots, s_2.$$

Thus y_1, y_2 show evidence for $\psi > 0$ if y_1 is too close to $y_1 + y_2$.

See also Example 4.40. ■

Example 7.35 (Permutation test) Let $Y_1, \dots, Y_m$ and $Y_{m+1}, \dots, Y_n$ be independent random samples with densities $g(y)$ and $g(y - \theta)$, where g is unknown. One possibility here is to base a test of $\theta = 0$ on the two-sample t statistic

$$T = \frac{\bar{Y}_2 - \bar{Y}_1}{\left[\left(\frac{1}{m} + \frac{1}{n-m} \right) \{ (m-1)S_1^2 + (n-m)S_2^2 \} \right]^{1/2}},$$

where $\bar{Y}_2$ and S_2^2 are the average and variance of $Y_{m+1}, \dots, Y_n$ and $\bar{Y}_1$ and S_1^2 are the corresponding quantities for $Y_1, \dots, Y_m$.

Under the null hypothesis $Y_1, \dots, Y_n$ form a random sample with unknown density g , and the set of order statistics $Y_{(1)}, \dots, Y_{(n)}$ is a minimal sufficient statistic. The conditional null distribution of $Y_1, \dots, Y_n$ given the observed values $y_{(1)}, \dots, y_{(n)}$ of the order statistics puts equal mass on each of the $n!$ permutations of $y_1, \dots, y_n$, so the conditional P-value is

$$\Pr_0(T \geq t_{\text{obs}} | Y_{(1)}, \dots, Y_{(n)}) = \frac{1}{n!} \sum H\{t(y_{\text{perm}}) \geq t_{\text{obs}}\}$$

where the sum is over all permutations y_{perm} of $y_1, \dots, y_n$. ■

Invariance

Section 5.3 describes models in which data y were transformed by the action of a group $\mathcal{G}$ on the sample space, thereby inducing a similar group action on the parameter space. In many cases it is appropriate that tests be invariant to the subgroup $\mathcal{G}_0$ of such transformations that preserves the null hypothesis. When testing the hypothesis $\mu = 0$ for a sample y from the $N(\mu, \sigma^2)$ distribution, for example, we might seek a test that is unaffected by replacing y by τy . The corresponding parameter transformation maps σ^2 to $\tau^2 \sigma^2$, thereby preserving the null hypothesis. To see some consequences of

requiring such invariances, suppose that the null hypothesis splits the parameter space Θ into disjoint parts Θ_0 and Θ_1 corresponding to the null and alternative hypotheses. The problem is then said to be invariant under $\mathcal{G}_0$ if

$$\Pr\{g(Y) \in \mathcal{A}; \theta\} = \Pr\{Y \in \mathcal{A}; g^*(\theta)\}$$

for all subsets $\mathcal{A}$ of the sample space and all $g \in \mathcal{G}_0$ and corresponding $g^* \in \mathcal{G}_0^*$, where g^* satisfies $g^*(\Theta) = \Theta$, $g^*(\Theta_0) = \Theta_0$ and $g^*(\Theta_1) = \Theta_1$. Thus the action of $\mathcal{G}_0^*$ on Θ leaves Θ_0 and Θ_1 unchanged: whatever transformation is applied to Y , the null hypothesis remains equally true or false. Hence the evidence for or against the hypotheses is unaffected by observing $g(Y)$ rather than Y , for any $g \in \mathcal{G}_0$. A test with critical region $\mathcal{V}_\alpha$ is then said to be *invariant* if

$$Y \in \mathcal{V}_\alpha \text{ if and only if } g(Y) \in \mathcal{V}_\alpha \text{ for all } g \in \mathcal{G}_0, \quad (7.35)$$

implying that its properties are unaffected by transformation. The hope is that appeal to invariance will simplify the problem by eliminating nuisance parameters. We can then search among invariant tests for one with high power or other good properties. As every invariant statistic is a function of a maximal invariant, we start by seeking a maximal invariant under $\mathcal{G}_0$.

Example 7.36 (Student t test) Suppose that we wish to test $\mu = \mu_0$ against the alternative $\mu \neq \mu_0$, based on a normal random sample $Y_1, \dots, Y_n$, with no restriction on the variance σ^2 . We take $\theta = (\mu, \sigma)$, so Θ_0 is $\{\mu_0\} \times \mathbb{R}_+$ and

$$\Theta_1 = \{(-\infty, \mu_0) \cup (\mu_0, \infty)\} \times \mathbb{R}_+.$$

Let $V = (n-1)^{-1} \sum (Y_j - \bar{Y})^2$. The statistic $(\bar{Y}, V^{1/2})$ is minimal sufficient in the full model and can form the basis of our discussion. As $(\bar{Y}, V^{1/2})$ takes values in the parameter space Θ , Example 5.21 implies that an element $g_{(\eta, \tau)}$ of the group $\mathcal{G}^*$ acting on Θ transforms $(\bar{Y}, V^{1/2})$ to $(\eta + \tau \bar{Y}, \tau V^{1/2})$. This reduction to a minimal sufficient statistic taking values in Θ means that our discussion below may be expressed in terms of $\mathcal{G}^*$ rather than the group $\mathcal{G}$ acting on the original data Y .

The subset of $\mathcal{G}^*$ that preserves Θ_0 must have

$$g_{(\eta, \tau)}(\mu_0, \sigma) = (\eta + \tau \mu_0, \tau \sigma) = (\mu_0, a)$$

for some $a > 0$, and this implies that $\eta = \mu_0 - \tau \mu_0$ but imposes no restriction on τ . Hence the largest such subset is

$$\mathcal{G}_0^* = \{g_{(\mu_0 - \tau \mu_0, \tau)} : \tau > 0\}.$$

To verify that $\mathcal{G}_0^*$ is a subgroup of $\mathcal{G}^*$, note that it is closed, because

$$g_{(\mu_0 - \tau \mu_0, \tau)} \circ g_{(\mu_0 - \sigma \mu_0, \sigma)} = g_{(\mu_0 - \tau \mu_0 + \tau(\mu_0 - \sigma \mu_0), \tau \sigma)} = g_{(\mu_0 - \tau \sigma \mu_0, \tau \sigma)}$$

is also an element of $\mathcal{G}_0^*$, that setting $\tau = 1$ gives the identity element $g_{(0, 1)}$, and that $g_{(\mu_0 - \tau \mu_0, \tau)}$ has inverse $g_{(\mu_0 - \tau^{-1} \mu_0, \tau^{-1})}$ also an element of $\mathcal{G}_0^*$. Moreover $\mathcal{G}_0^*$ preserves Θ_1 , because if $\mu \neq \mu_0$, then

$$g_{(\mu_0 - \tau \mu_0, \tau)}(\mu, \sigma) = (\mu_0 - \tau \mu_0 + \tau \mu, \tau \sigma) = (\mu_0 + \tau(\mu - \mu_0), \tau \sigma) \in \Theta_1.$$

Now $g_{(\mu_0 - \tau\mu_0, \tau)}$ maps the Student t pivot $T(\mu_0) = n^{1/2}(\bar{Y} - \mu_0)/V^{1/2}$ to

$$n^{1/2} \frac{\mu_0 - \tau\mu_0 + \tau\bar{Y} - \mu_0}{\tau V^{1/2}} = n^{1/2} \frac{\tau(\bar{Y} - \mu_0)}{\tau V^{1/2}} = T(\mu_0),$$

so $T(\mu_0)$ is invariant under $\mathcal{G}_0$. To verify that it is a maximal invariant, we find an estimator that lies in Θ_0 and is equivariant under $\mathcal{G}_0^*$, such as $s(\bar{Y}, V^{1/2}) = (\mu_0, V^{1/2})$. Then a maximal invariant is (page 185)

$$\begin{aligned} g_{(\mu_0 - \mu_0 V^{1/2}, V^{1/2})}^*(\bar{Y}, V^{1/2}) &= g_{(\mu_0 - \mu_0 V^{-1/2}, V^{-1/2})}^*(\bar{Y}, V^{1/2}) \\ &= (\mu_0 - \mu_0 V^{-1/2} + V^{-1/2} \bar{Y}, V^{-1/2} V^{1/2}) \\ &= (\mu_0 + (\bar{Y} - \mu_0) V^{-1/2}, 1), \end{aligned}$$

the second component of which can obviously be discarded. Under the null hypothesis μ_0 is known, so $T(\mu_0)$ is also maximal invariant, as we had anticipated. Hence any critical region based on $T(\mu_0)$ would be unaltered if a sample y was replaced by $\mu_0 - \tau\mu_0 + \tau y$, for any $\tau > 0$, because

$$n^{1/2} \frac{\bar{y} - \mu_0}{v^{1/2}} \in \mathcal{A} \quad \text{if and only if} \quad n^{1/2} \frac{\mu_0 - \tau\mu_0 + \tau\bar{y} - \mu_0}{\tau v^{1/2}} \in \mathcal{A}$$

for any set $\mathcal{A} \subset \mathbb{R}$, thus verifying (7.35). Thus any critical region based on $T(\mu_0)$ is invariant. An example is

$$\left\{ (y_1, \dots, y_n) : n^{1/2} \left| \frac{\bar{y} - \mu_0}{v^{1/2}} \right| \geq t_{n-1}(1 - \alpha) \right\},$$

which has size 2α and is uniformly most powerful unbiased against two-sided alternatives, in addition to being invariant. ■

7.3.4 Link with confidence intervals

There is a close link between tests and the construction of confidence intervals. If the density of Y depends on a scalar parameter θ , we define a *level α upper confidence limit* to be a function $T^\alpha = t^\alpha(Y)$ of Y such that

$$\Pr(\theta \leq T^\alpha; \theta) = 1 - \alpha \quad \text{for all } \theta, \quad (7.36)$$

and that $T^{\alpha_1} \leq T^{\alpha_2}$ whenever $\alpha_1 > \alpha_2$. This requirement is similar to the nesting of critical regions for tests and is imposed for the same reasons of consistency; it implies that T^α is non-increasing in α . Lower confidence limits may be defined analogously.

The random quantity in (7.36) is T^α . An equi-tailed $(1 - 2\alpha)$ confidence interval for θ is $(T^{1-\alpha}, T^\alpha)$. If the reparametrization $\psi = \psi(\theta)$ is monotonic increasing, then $\psi(T^\alpha)$ is an upper confidence limit for ψ .

In many cases confidence limits are derived from a pivot $Z(\theta)$, a function of the data and θ with the same distribution for all θ . If this distribution is continuous, we can find a z_α such that

$$\Pr\{Z(\theta) \leq z_\alpha; \theta\} = \alpha \quad \text{for all } \theta.$$

$t_{n-1}(\alpha)$ is the α quantile of the t_{n-1} distribution.

If $Z(\theta)$ is decreasing in θ for every possible value of Y , then the solution in θ to the equation $Z(\theta) = z_\alpha$ can be taken as an upper $(1 - \alpha)$ confidence limit for θ . We applied this argument to approximate normal pivots and the signed likelihood ratio statistic in Sections 3.1.1 and 4.5.2; see Figures 3.1 and 4.7.

Now suppose that $\mathcal{Y}_\alpha(\theta_0)$ is a critical region of size α constructed for tests of $\theta = \theta_0$ against lower alternatives $\theta < \theta_0$. As θ_0 increases, the critical region will vary and we can define the set

$$\{\theta : Y \notin \mathcal{Y}_\alpha(\theta)\}$$

of values of θ not rejected by the test and hence compatible with the data at level α . Under natural monotonicity conditions the supremum of this set can be taken as an upper $(1 - \alpha)$ confidence limit T^α . This inversion of a collection of critical regions to obtain a confidence interval allows us to use good tests to construct good confidence intervals. For example, the Neyman–Pearson lemma tells us that uniformly most powerful tests of simple hypotheses are commonly based on likelihood ratio statistics, which will therefore also be the basis for shortest confidence intervals.

In many cases we can express the above argument as follows. Let $G(t; \theta_0)$ denote the null distribution function of a continuous test statistic T when the null hypothesis is $\theta = \theta_0$. Then the P-value

$$p_{\text{obs}}(\theta_0) = \Pr_0(T \geq t_{\text{obs}}) = 1 - G(t_{\text{obs}}; \theta_0)$$

is a realization of $P(\theta_0) = 1 - G(T; \theta_0)$, and the probability integral transform (Section 2.3) implies that the null distribution of $P(\theta_0)$ is uniform on $(0, 1)$. If the test rejects when $P(\theta_0) < \alpha$, then the set $\{\theta : \alpha \leq P(\theta)\}$ is a one-sided $(1 - \alpha)$ confidence set. In the two-sided case we take $\{\theta : \alpha \leq P(\theta) \leq 1 - \alpha\}$.

This argument applies when we can eliminate parameters other than θ by appeal to similarity or invariance; otherwise it can be sometimes be applied approximately, as with the likelihood ratio statistic. Minor complications arise when T is discrete; see Example 7.38.

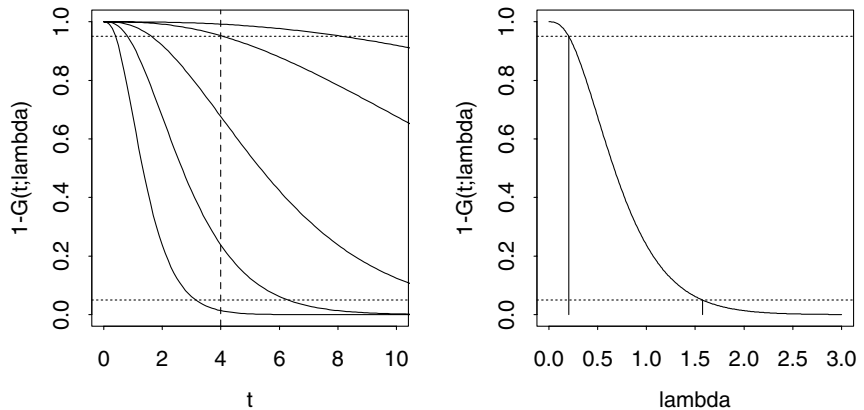
Example 7.37 (Exponential density) Let $Y_1, \dots, Y_n$ be a random sample from the exponential density with parameter λ , and let a test of $\lambda = \lambda_0$ be conducted against the two-sided alternative $\lambda \neq \lambda_0$. We saw in Example 7.22 that the null density of $T = \sum Y_j$ is gamma with shape parameter n and scale λ_0 , so the null hypothesis is rejected at level $(1 - 2\alpha)$ if

$$p_{\text{obs}}(\lambda_0) = \Pr_0(T \geq t_{\text{obs}}) = \int_{\lambda_0 t_{\text{obs}}}^{\infty} \frac{v^{n-1}}{\Gamma(n)} e^{-v} dv$$

lies outside the interval $(\alpha, 1 - \alpha)$. For a given value of t_{obs} , this probability depends on λ_0 , as shown in Figure 7.7, and a $(1 - 2\alpha)$ confidence interval can be determined as the set of values of λ for which $\alpha \leq p_{\text{obs}}(\lambda) \leq 1 - \alpha$. ■

The interpretation of two-sided confidence intervals as providing random upper and lower bounds is direct and useful for scalar parameters. Confidence regions for vector θ require a shape. It is natural to base this on likelihood, insisting that a confidence

Figure 7.7 Inversion of a two-sided test with level 0.9 to form confidence interval. Left: significance levels $p_{\text{obs}}(\lambda_0)$ for $\lambda_0 = 0.1, 0.2, 0.5, 1, 2$ (top to bottom). Horizontal lines show probabilities 0.05, 0.95 and the vertical line shows $t_{\text{obs}} = 4$. Hypotheses $\lambda_0 = 2, 0.1$ are rejected, hypotheses $\lambda_0 = 1, 0.5$ are not rejected, and $\lambda_0 = 0.2$ is just rejected. Right: significance level $p_{\text{obs}}(\lambda)$ as a function of λ . Values of λ for which $0.05 \leq p_{\text{obs}}(\lambda) \leq 0.95$ are contained in the 0.9 confidence interval.



region $\mathcal{R}_\alpha$ be such that $\Pr(\theta \in \mathcal{R}_\alpha; \theta) = \alpha$ for all θ and that $L(\theta) \geq L(\theta')$ for any $\theta \in \mathcal{R}_\alpha$ and $\theta' \notin \mathcal{R}_\alpha$. This amounts to computing $\mathcal{R}_\alpha$ by inverting the likelihood ratio statistic, typically using its asymptotic distribution, perhaps with Bartlett adjustment.

Often the test inverted to obtain limits of confidence intervals is not exact. Then there is *coverage error*, defined as the difference between the actual and nominal probabilities that the confidence set contains the parameter,

$$\Pr(T^{\alpha_1} < \theta \leq T^{\alpha_2}; \theta) - (\alpha_1 - \alpha_2), \quad \text{for } \alpha_1 > \alpha_2. \quad (7.37)$$

It can be helpful to know where the error occurs. The limit T^α is said to be *conservative* if it tends to be too high, that is, $\Pr(\theta \leq T^\alpha; \theta) \geq 1 - \alpha$; confidence intervals for which (7.37) is positive are called conservative.

Example 7.38 (Binomial density) An equitailed $(1 - 2\alpha)$ confidence interval for the probability π of a binomial variable Y with denominator m may be found in various ways. Exact limits may be found by inverting tests based on Y . Having observed $Y = y$, the significance level for testing the null hypothesis $\pi = \pi_0$ against the one-sided alternative $\pi < \pi_0$ is

$$\Pr_0(Y \leq y) = \Pr(Y \leq y; \pi_0) = \sum_{r=0}^y \binom{m}{r} \pi_0^r (1 - \pi_0)^{m-r},$$

so the upper α limit π^α is the solution to

$$\Pr(Y \leq y; \pi) = \sum_{r=0}^y \binom{m}{r} \pi^r (1 - \pi)^{m-r} = \alpha,$$

and equals 1 if $y = m$. A similar argument with alternative $\pi > \pi_0$ shows that the lower α limit π_α is the solution to

$$\Pr(Y \geq y; \pi) = \sum_{r=y}^m \binom{m}{r} \pi^r (1 - \pi)^{m-r} = \alpha,$$

Otherwise they are called liberal.

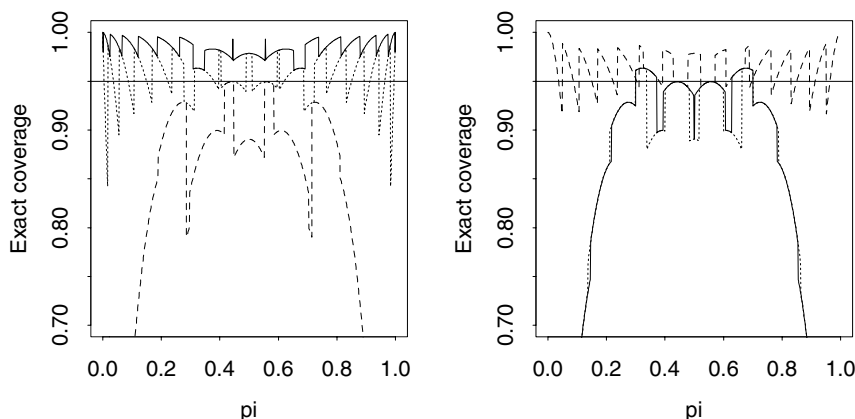


Figure 7.8 Exact coverages of equi-tailed 0.95 confidence intervals for the binomial parameter π , as functions of π , when $m = 10$. The horizontal line shows the target coverage. Left: exact (solid), score (dots) and maximum likelihood estimator (dashes). Right: signed likelihood ratio statistic (solid), modified signed likelihood ratio statistic (dots) and modified maximum likelihood estimator (dashes), obtained by replacing m and r by $m + 2$ and $r + 1$ (dashes).

but equals 0 if $y = 0$. It turns out that π^α and π_α are expressible using quantiles of the F distribution, giving $(1 - 2\alpha)$ confidence interval

$$\left(\left\{ 1 + \frac{m - y + 1}{y F_{2y, 2(m-y+1)}^{-1}(\alpha)} \right\}^{-1}, \left\{ 1 + \frac{m - y}{(y + 1) F_{2(y+1), 2(m-y)}^{-1}(1 - \alpha)} \right\}^{-1} \right),$$

with the changes mentioned above when $y = 0$ or $y = m$. This interval is exact in the sense that no approximation of binomial probabilities is involved.

Approximate intervals can be based on asymptotic standard normal distributions of the score statistic, the maximum likelihood estimator $\hat{\pi} = Y/m$ or the signed likelihood ratio statistic,

$$Z_1(\pi) = (Y - m\pi) / \{m\pi(1 - \pi)\}^{1/2},$$

$$Z_2(\pi) = (\hat{\pi} - \pi) / \{\hat{\pi}(1 - \hat{\pi})/m\}^{1/2},$$

$$Z_3(\pi) = \text{sign}(\hat{\pi} - \pi) \left(2 \left[Y \log(\hat{\pi}/\pi) + (m - Y) \log \{(1 - \hat{\pi})/(1 - \pi)\} \right] \right)^{1/2},$$

as well as on a quantity $Z^*(\pi) = Z_3(\pi) + Z_3(\pi)^{-1} \log \{Z_2(\pi)/Z_3(\pi)\}$ motivated in Section 12.3.3. The confidence interval based on each of these is the set of π for which $|Z(\pi)| < z_{1-\alpha}$; this must be found numerically for $Z_3(\pi)$ and $Z^*(\pi)$. Any of these intervals has coverage $\sum_{y=0}^m \binom{m}{y} \pi^y (1 - \pi)^{m-y} I_{1-2\alpha}(\pi, y)$, where $I_{1-2\alpha}(y, \pi)$ indicates that π lies in an interval of nominal level $(1 - 2\alpha)$ based on y .

Figure 7.8 compares the coverages for $\alpha = 0.025$ and $m = 10$. That of the exact interval always exceeds 0.975, so it is quite conservative, while that of the interval based on $Z_1(\pi)$ is fairly close to its nominal level overall. Intervals based on $Z_2(\pi)$ undercover for most π . The intervals based on $Z_3(\pi)$ and $Z^*(\pi)$ have coverage close to nominal for $0.3 < \pi < 0.7$, while perhaps the best overall performance is obtained from $Z_2(\pi)$ with m and y replaced by $m + 2$ and $y + 1$. ■

This example suggests that in highly discrete situations approximate confidence intervals may be preferable to exact ones. Moreover exact tests will inherit the conservatism and tend to reject too rarely. The difference decreases as the sample size increases, but even with $m = 50$ the mean exact coverage is about 0.97 in the binomial case.

$F_{v_1, v_2}(y)$ is the distribution function of an F variable with v_1, v_2 degrees of freedom.

Exercises 7.3

- 1 Show that (7.26) has mean and variance roughly p_{obs} and $p_{\text{obs}}(1 - p_{\text{obs}})/R$. Hence give minimum values of R for obtaining 5% relative error in estimation of $p_{\text{obs}} = 0.5, 0.2, 0.1, 0.05, 0.01, 0.001$. Discuss.
- 2 In Example 7.22, calculate the significance level for testing $H_0 : \lambda = 1$ against $H_1 : \lambda = 4$, based on the data 1.2, 3, 1.5, 0.3.
- 3 If $U \sim U(0, 1)$, show that $\min(U, 1 - U) \sim U(0, \frac{1}{2})$. Hence justify the computation of a two-sided significance level as $2 \min(P^-, P^+)$.
- 4 Consider testing the hypothesis that $\mu = \mu_0$ based on a random sample $Y_1, \dots, Y_n$ from the $N(\mu, \sigma^2)$ distribution, with two-sided alternative $\mu \neq \mu_0$. Show that the power of the region (7.31) is $\Phi(z_{\alpha/2} + \delta) + \Phi(z_{\alpha/2} - \delta)$, where $\delta = n^{1/2}(\mu - \mu_0)/\sigma$. Sketch this as a function of δ for $\alpha = 0.025$, and explain why it is invariant to the sign of $\mu - \mu_0$.
- 5 Check the power calculation for the sign test in Example 7.30.
- 6 Consider testing the hypothesis that a binomial random variable has probability $\pi = 1/2$ against the alternative that $\pi > 1/2$. For what values of α does a uniformly most powerful test exist when the denominator is $m = 5$?
- 7 In a random sample $Y_1, \dots, Y_n$ from the gamma density with shape κ and scale λ , find a locally most powerful test of the null hypothesis $\kappa = 1$.
- 8 If I is Bernoulli with probability $p = (\alpha_2 - \alpha)/(\alpha_2 - \alpha_1)$ and $\mathcal{Y}_{\alpha_1}$ and $\mathcal{Y}_{\alpha_2}$ are critical regions of sizes α_1, α_2 , show that the critical region $\mathcal{Y} = I\mathcal{Y}_{\alpha_1} + (1 - I)\mathcal{Y}_{\alpha_2}$ has size α .
- 9 Y_1, Y_2 are independent gamma variables with known shape parameters ν_1, ν_2 and scale parameters λ_1, λ_2 , and it is desired to test the null hypothesis H_0 that $\lambda_1 = \lambda_2 = \lambda$, with λ unknown. Show that a minimal sufficient statistic for λ under H_0 is $Y_1 + Y_2$, find its distribution, and show that it is complete. Hence show that the test is based on the conditional distribution of Y_1 given $Y_1 + Y_2$ and that significance levels are computed from integrals of form

$$\frac{\Gamma(\nu_1 + \nu_2)}{\Gamma(\nu_1)\Gamma(\nu_2)} \int_0^{y_1/(y_1+y_2)} u^{\nu_1-1}(1-u)^{\nu_2-1} du.$$

Explain how this argument is useful in comparison of the scale parameters of two independent exponential samples.

- 10 Independent data pairs $(X_1, Z_1), \dots, (X_n, Z_n)$ arise from a joint density $f(x, z)$. The null hypothesis is that X and Z are independent, so $f(x, z) = g(x)h(z)$ for some unknown densities g and h and all x and z . Show that the order statistics $X_{(1)}, \dots, X_{(n)}$ and $Z_{(1)}, \dots, Z_{(n)}$ are minimal sufficient for g and h under the null hypothesis, and deduce that a similar test has P-value

$$p_{\text{obs}} = \frac{1}{n!} \sum H\{t(y_{\text{perm}}) \geq t_{\text{obs}}\},$$

where the sum is over all $y_{\text{perm}} = \{(x_1, z_{\pi(1)}), \dots, (x_n, z_{\pi(n)})\}$ with the observed values of the z s permuted, the x s being held fixed.

If the test statistic is $T = (n^{-1} \sum X_j Z_j - \bar{X} \bar{Z}) / (S_X^2 S_Z^2)^{1/2}$, S_X^2 and S_Z^2 being the sample variances of the X_j and the Z_j , show that it is equivalent to base the test on $\sum X_j Z_j$.

- 11 In a scale family, $Y = \tau \varepsilon$, where ε has a known density and $\tau > 0$. Consider testing the null hypothesis $\tau = \tau_0$ against the alternative $\tau \neq \tau_0$. Show that the appropriate group for constructing an invariant test has just one element (apart from permutations) and hence show that the test may be based on the maximal invariant $Y_{(1)}/\tau_0, \dots, Y_{(n)}/\tau_0$. When ε is exponential, show that the invariant test is based on $\bar{Y}/\tau_0$.
- 12 One natural transformation of a binomial variable R is reversal of ‘success’ and ‘failure’. Show that this maps R to $m - R$, where m is the denominator, and that

the induced transformation on the parameter space maps π to $1 - \pi$. Which of the critical regions (a) $\mathcal{Y}_1 = \{0, 1, 20\}$, (b) $\mathcal{Y}_2 = \{0, 1, 19, 20\}$, (c) $\mathcal{Y}_3 = \{0, 1, 10, 19, 20\}$, (d) $\mathcal{Y}_4 = \{8, 9, 10, 11, 12\}$, is invariant for testing $\pi = \frac{1}{2}$ when $m = 20$? Which is preferable and why?

- 13 The incidence of a rare disease seems to be increasing. In successive years the numbers of new cases have been $y_1, \dots, y_n$. These may be assumed to be independent observations from Poisson distributions with means $\lambda\theta, \dots, \lambda\theta^n$. Show that there is a family of tests each of which, for any given value of λ , is a uniformly most powerful test of its size for testing $\theta = 1$ against $\theta > 1$.
- 14 A random sample $Y_1, \dots, Y_n$ is available from the *Type I Pareto* distribution

$$F(y; \psi) = \begin{cases} 1 - y^{-\psi}, & y \geq 1, \\ 0, & y < 1. \end{cases}$$

Find the likelihood ratio statistic to test that $\psi = \psi_0$ against $\psi = \psi_1$, where ψ_0, ψ_1 are known, and show how to calculate a P-value when $\psi_0 > \psi_1$.
How does your answer change if the distribution is

$$F(y; \psi, \lambda) = \begin{cases} 1 - (y/\lambda)^{-\psi}, & y \geq \lambda, \\ 0, & y < \lambda, \end{cases}$$

with $\lambda > 0$ unspecified?

7.4 Bibliographic Notes

The main concepts described in this chapter belong to the core of statistical theory and were developed in the first half of the twentieth century by Fisher, Neyman, Pearson and others; other treatments are contained in most books on mathematical statistics. See for example the treatments of estimation in Silvey (1970), Rice (1988), Casella and Berger (1990) and Bickel and Doksum (1977), or at a more advanced level Cox and Hinkley (1974), Lehmann (1983) and Shao (1999).

Kernel density estimation has been extensively studied since it was proposed in the 1950s. Among numerous excellent expositions are Silverman (1986), Scott (1992), Wand and Jones (1995), and Bowman and Azzalini (1997). The last of these is more practical in emphasis, while Wand and Jones (1995) contains a detailed discussion of the choice of bandwidth, a topic on which there has been much progress in the 1990s. Although cross-validation is an important paradigm for selection of bandwidths and related smoothing parameters in other non- and semi-parametric contexts, other approaches to bandwidth selection give better results; see Sheather and Jones (1991). Stone (1974) is a fundamental reference on cross-validation.

Estimators based on estimating functions are widely used in practice, but there are few general expositions of them at this level. Godambe (1991) is an interesting collection of papers on the topic, with many further references, while McLeish and Small (1994) give a more abstract treatment. A fundamental reference for the role of the influence function in robust statistics is Hampel *et al.* (1986). Inference for stochastic processes is discussed in books by Hall and Heyde (1980), Basawa and Scott (1981), and Guttorp (1991), while Sørensen (1999) reviews the asymptotic theory for estimating functions.

Although the idea of significance testing goes back hundreds of years, the development of underlying theory is more recent. R. A. Fisher made extensive informal use of P-values, but resisted what he saw as the over-formalization due to Neyman and E. S. Pearson. They introduced the idea of testing as a choice between two hypotheses and introduced the notions of size, power and so forth in work that prefigured the later development of decision theory. Their joint papers are collected in Neyman and Pearson (1967). The theory of testing is explained more fully in Lehmann (1983) and in Chapters 3–6 of Cox and Hinkley (1974). Bartlett correction was first described by Bartlett (1937). Example 7.38 is based on Agresti and Coull (1998), Agresti and Caffo (2000), and Greenland (2001).

7.5 Problems

- 1 In Example 7.2 show that $\hat{\psi} \stackrel{D}{=} \exp\{\mu + \sigma n^{-1/2}Z + \sigma^2 V/(2n)\}$. Hence give an explicit expression for $E(\hat{\psi}^r)$ and compute the analogue of Table 7.1. Discuss your results.
- 2 Let $Y_1, \dots, Y_n$ be a random sample from an unknown density f . Let I_j indicate whether or not Y_j lies in the interval $(a - \frac{1}{2}h, a + \frac{1}{2}h]$, and consider $R = \sum I_j$. Show that R has a binomial distribution with denominator n and probability

$$\int_{a-\frac{1}{2}h}^{a+\frac{1}{2}h} f(y) dy.$$

Hence show that $R/(nh)$ has approximate mean and variance $f(a) + \frac{1}{2}h^2 f''(a)$ and $f(a)/nh$, where f'' is the second derivative of f .

What implications have these results for using the histogram to estimate $f(a)$?

- 3 Suppose that the random variables $Y_1, \dots, Y_n$ are such that

$$E(Y_j) = \mu, \quad \text{var}(Y_j) = \sigma_j^2, \quad \text{cov}(Y_j, Y_k) = 0, \quad j \neq k,$$

where μ is unknown and the σ_j^2 are known. Show that the linear combination of the Y_j 's giving an unbiased estimator of μ with minimum variance is

$$\sum_{j=1}^n \sigma_j^{-2} Y_j / \sum_{j=1}^n \sigma_j^{-2}.$$

Suppose now that Y_j is normally distributed with mean βx_j and unit variance, and that the Y_j are independent, with β an unknown parameter and the x_j known constants. Which of the estimators

$$T_1 = n^{-1} \sum_{j=1}^n Y_j / x_j, \quad T_2 = \sum_{j=1}^n Y_j x_j / \sum_{j=1}^n x_j^2$$

is preferable and why?

- 4 In n independent food samples the bacterial counts $Y_1, \dots, Y_n$ are presumed to be Poisson random variables with mean θ . It is required to estimate the probability that a given sample would be uncontaminated, $\pi = \Pr(Y_j = 0)$. Show that $U = n^{-1} \sum I(Y_j = 0)$, the proportion of the samples uncontaminated, is unbiased for π , and find its variance. Using the Rao–Blackwell theorem or otherwise, show that an unbiased estimator of π having smaller variance than U is $V = \{(n-1)/n\}^{n\bar{Y}}$, where $\bar{Y} = n^{-1} \sum Y_j$. Is this a minimum variance unbiased estimator of π ? Find $\text{var}(V)$ and hence give the asymptotic efficiency of U relative to V .

- 5 Let $Y_1, \dots, Y_n$ be independent Poisson variables with means $x_1\beta, \dots, x_n\beta$, where $\beta > 0$ is an unknown scalar and the $x_j > 0$ are known scalars. Show that $T = \sum Y_j x_j / \sum x_j^2$ is an unbiased estimator of β and find its variance. Find a minimal sufficient statistic S for β , and show that the conditional distribution of Y_j given that $S = s$ is multinomial with mean $s x_j / \sum_i x_i$. Hence find the minimum variance unbiased estimator of β . Is it unique?
- 6 Given that there is a 1–1 mapping between $x_1 < \dots < x_n$ and the sums $s_1, \dots, s_n$, where $s_r = \sum x_j^r$, show that the order statistics of a random sample form a complete minimal sufficient statistic in the class of all continuous densities. You may find it useful to consider the exponential family density

$$f(y; \theta) \propto \exp(-x^{2n} + \theta_1 x + \dots + \theta_n x^n).$$

- 7 Find the maximum likelihood estimator of β based on a random sample from the shifted exponential density $f(y) = e^{-(y-\beta)}$ for $y \geq \beta$. Show that $\hat{\beta}$ is biased but consistent. Does it satisfy the Cramér–Rao lower bound?
- 8 (a) Let $Y_1, \dots, Y_n$ be a random sample from the exponential density $\lambda e^{-\lambda y}$, $y > 0$, $\lambda > 0$. Say why an unbiased estimator W for λ should have form a/S , and hence find a . Find the Fisher information for λ and show that $E(W^2) = (n-1)\lambda^2/(n-2)$. Deduce that no unbiased estimator of λ attains the Cramér–Rao lower bound, although W does so asymptotically.
- (b) Let $\psi = \Pr(Y > a) = e^{-\lambda a}$, for some constant a . Show that

$$I(Y_1 > a) = \begin{cases} 1, & Y_1 > a, \\ 0, & \text{otherwise,} \end{cases}$$

is an unbiased estimator of ψ , and hence obtain the minimum variance unbiased estimator. Does this attain the Cramér–Rao lower bound for ψ ?

- 9 Let $X_1, \dots, X_n$ represent the times of the first n events in a Poisson process of rate μ^{-1} observed from time zero; thus $0 < X_1 < \dots < X_n$. Show that $W = 2(X_1 + \dots + X_n)/\{n(n+1)\}$ is an unbiased estimator of μ , and establish that its Rao–Blackwellized form is $T = X_n/n$. Find $\text{var}(W)$ and give the asymptotic efficiency of W relative to T .
- 10 Show that no unbiased estimator exists of $\psi = \log\{\pi/(1-\pi)\}$, based on a binomial variable with probability π .
- 11 Let $Y_j = \eta + \tau \varepsilon_j$, where $\varepsilon_1, \dots, \varepsilon_n$ is a random sample from a known density. Show that the set of order statistics $Y_{(1)}, \dots, Y_{(n)}$ is in general minimal sufficient for η, τ (Example 4.12). By considering $(Y_{(2)} - Y_{(1)})/(Y_{(n)} - Y_{(1)})$ show that it is not complete.
- 12 Show that when the data are normal, the efficiency of the Huber estimating function $g_c(y; \theta)$ compared to the optimal function $g_\infty(y; \theta)$ is

$$\frac{\{1 - 2\Phi(-c)\}^2}{1 + 2\{c^2\Phi(-c) - \Phi(-c) - c\phi(c)\}}.$$

Hence verify that the efficiency is 0.95 when $c = 1.345$.

- 13 Compare the performance of the estimating function

$$g(y; \theta) = \begin{cases} y - \theta, & |y - \theta| < c, \\ 0, & \text{otherwise,} \end{cases}$$

with that of the Huber function $g_c(y; \theta)$ for the distributions in Example 7.19.

- 14 Show how (a) the Poisson birth process in Example 4.6, and (b) the Markov chain likelihood in Section 6.1.1, fall into the framework for dependent data outlined in Section 7.2.3.
- 15 Let $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, with both parameters unknown. Suppose that we wish to test $\mu = \mu_0$ against the one-sided alternative $\mu > \mu_0$. By considering separately the cases

$\bar{Y} \geq \mu_0$ and $\bar{Y} < \mu_0$, show that the likelihood ratio statistic is

$$W_p(\mu_0) = \begin{cases} n \log \left\{ 1 + \frac{T(\mu_0)^2}{n-1} \right\}, & \bar{Y} \geq \mu_0, \\ 0, & \bar{Y} < \mu_0. \end{cases}$$

Hence justify the one-tailed significance level described in Example 7.25.

- 16 Independent random samples $Y_{i1}, \dots, Y_{in_i}$, where $n_i \geq 2$, are drawn from each of k normal distributions with means $\mu_1, \dots, \mu_k$ and common unknown variance σ^2 . Derive the likelihood ratio statistic W_p for the null hypothesis that the μ_i all equal an unknown μ , and show that it is a monotone function of

$$R = \frac{\sum_{i=1}^k n_i (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2}{\sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i\cdot})^2},$$

where $\bar{Y}_{i\cdot} = n_i^{-1} \sum_j Y_{ij}$ and $\bar{Y}_{\cdot\cdot} = (\sum n_i)^{-1} \sum_{i,j} Y_{ij}$. What is the null distribution of R ?

- 17 Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent random samples from continuous distributions F_X and F_Y . We wish to test the hypothesis H_0 that $F_X = F_Y$.

Define indicator variables $I_{ij} = I(X_i < Y_j)$ for $i = 1, \dots, m$, $j = 1, \dots, n$ and let $U = \sum_{i,j} I_{ij}$. Assuming that H_0 is true, (i) show that $E(U) = mn/2$; (ii) find $\text{cov}(I_{ij}, I_{ik})$ and $\text{cov}(I_{ij}, I_{kl})$, where i, j, k, l are distinct; and (iii) hence show that $\text{var}(U) = mn(m+n+1)/12$. Why is it important that the underlying distributions are continuous?

Here are the weight gains (gms) of rats fed on low and high protein diets:

High	83	97	104	107	113	119	123	124	129	134	146	161
Low	70	85	94	101	106	118	132					

Use the approximate normality of U to test for a difference between diets.

- 18 Below are diastolic blood pressures (mm Hg) of ten patients before and after treatment for high blood pressure. Test the hypothesis that the treatment has no effect on blood pressure using a Wilcoxon signed-rank test, (a) using the exact significance level and (b) using a normal approximation. Discuss briefly.

Before	94	105	101	106	118	107	96	102	114	95
After	96	96	95	103	105	111	86	90	107	84

- 19 (a) A random sample of size $n = 2$ is taken from $f(y)$. For $0 < \alpha < 1/2$, find a critical region of size α for testing that $f(y)$ is

$$f_0(y) = \begin{cases} \theta^{-1}, & 0 < y < \theta, \\ 0, & \text{otherwise,} \end{cases}$$

when $\theta = 1$, against the alternative that $f(y)$ is the exponential density $f_1(y) = e^{-y}$, $y > 0$. Is there a best critical region for testing $f = f_0$ against the composite hypothesis $f(y) = \lambda \exp(-\lambda y)$, $y > 0$, for some $\lambda > 0$?

(b) Show there is no best critical region when θ is unknown.

(c) Show that the largest order statistic $Y_{(2)}$ is sufficient for θ under the null model, and deduce that there is a uniformly most powerful test based on the ratio of conditional densities of Y given $Y_{(2)}$ under the two hypotheses. Show that the most powerful conditional critical region of size α is $\mathcal{Y}_\alpha = \{(y_1, y_2) : 0 \leq y_{(1)} \leq \alpha y_{(2)}\}$.

(d) Find the conditional critical region for general n .

- 20 If

$$f(x; \theta) = \begin{cases} \theta^\lambda \Gamma(\lambda)^{-1} x^{\lambda-1} e^{-\theta x}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where λ is known and θ is positive, deduce that there exists a uniformly most powerful test of size α of the hypothesis $\theta = \theta_0$ against the alternative $\theta > \theta_0$, and show that when $\lambda = 1/n$ the power function of the test is $1 - (1 - \alpha)^{\theta/\theta_0}$.

- 21 A source at location $x = 0$ pollutes the environment. Are cases of a rare disease $\mathcal{D}$ later observed at positions $x_1, \dots, x_n$ linked to the source?

Cases of another rare disease $\mathcal{D}'$ known to be unrelated to the pollutant but with the same susceptible population as $\mathcal{D}$ are observed at $x'_1, \dots, x'_m$. If the probabilities of contracting $\mathcal{D}$ and $\mathcal{D}'$ are respectively $\psi(x)$ and ψ' , and the population of susceptible individuals has density $\lambda(x)$, show that the probability of $\mathcal{D}$ at x , given that $\mathcal{D}$ or $\mathcal{D}'$ occurs there, is

$$\pi(x) = \frac{\psi(x)\lambda(x)}{\psi(x)\lambda(x) + \psi'\lambda(x)}.$$

Deduce that the probability of the observed configuration of diseased persons, conditional on their positions, is

$$\prod_{j=1}^n \pi(x_j) \prod_{i=1}^m \{1 - \pi(x'_i)\}.$$

The null hypothesis that $\mathcal{D}$ is unrelated to the pollutant asserts that $\psi(x)$ is independent of x . Show that in this case the unknown parameters may be eliminated by conditioning on having observed n cases of $\mathcal{D}$ out of a total $n + m$ cases. Deduce that the null probability of the observed pattern is $\binom{n+m}{n}^{-1}$.

If T is a statistic designed to detect decline of $\psi(x)$ with x , explain how permutation of case labels $\mathcal{D}, \mathcal{D}'$ may be used to obtain a significance level p_{obs} .

Such a test is typically only conducted after a suspicious pattern of cases of $\mathcal{D}$ has been observed. How will this influence p_{obs} ?